LOCAL LAPLACIAN ENERGY AND VERTEX-DELETION PROFILES OF GRAPHS

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Abstract. We study the local Laplacian energy $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)})$ of a vertex and the profile $\mathcal{M}_L(G) = \{\mathscr{LE}_G(v) : v \in V(G)\}$ of signed energy changes produced by vertex deletions. The global energy, the total local energy and the mean energy of the vertex-deleted deck satisfy one affine relation, and two graphs have the same profile exactly when their deletion decks agree after a rigid translation; a moment duality extends this to every central moment. Up to one translation, the profile is therefore the deletion deck. Unlike the adjacency energy, the Laplacian energy can increase when a vertex is deleted, because the deletion also moves the centering baseline. We prove that this can happen only at heavy vertices: $\mathscr{LE}_G(v) < 0$ forces $d_v > \overline{d}(G)/2$, while complete graphs, cycles and conference graphs show that the condition is not sufficient. Closed forms for complete multipartite, strongly regular, conference, cycle, corona and windmill graphs show that the local energy never exceeds $LE(G)$, with equality only at star centres, reaches $-\frac{n}{2}+O(1)$ on $K_{r,r}$, depends for strongly regular graphs only on the first subconstituent, and tends to $4/\pi$ on long cycles. We bound the profile variance through the degree modulus $\sqrt{4\sigma_d^2 + \overline{d}(G)^2}$, which is asymptotically sharp, and show that the profile separates Laplacian cospectral graphs exactly when their deletion decks differ; the smallest cospectral graphs with identical profiles have nine vertices. Finally, a census of all connected graphs through order ten shows that only 19 of the 24335 order-ten coincidences of LE and e_L extend to the full profile, each certified in exact arithmetic.

1. Introduction

Spectral graph theory investigates the relationship between a graph's structural characteristics and the eigenvalues and eigenvectors of associated matrices. Two fundamental matrices in this framework are the *adjacency matrix*, defined as $A(G) = [a_{ij}]_{i,j=1}^{n}$, and the diagonal *degree matrix*, $D(G) = \operatorname{diag}(d_1, d_2, \ldots, d_n)$, where

$$a_{ij} = \begin{cases} 1 & \text{if vertices } v_i \text{ and } v_j \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

The *Laplacian matrix* of a graph G is given by

$$L(G) = D(G) - A(G),$$

and it plays a central role in spectral graph theory due to its rich algebraic structure and its ability to capture combinatorial and topological properties of graphs [5, 6, 22]. Its eigenvalues, known collectively as the *Laplacian spectrum*, satisfy

$$0 = \lambda_1(L) \leq \lambda_2(L) \leq \cdots \leq \lambda_n(L),$$

with the multiplicity of zero equal to the number of connected components in the graph [2, 7]. The Laplacian spectrum has diverse applications across applied mathematics and network science, including network centrality [26], consensus and synchronization in dynamical systems [3, 17], and modeling diffusion and alliance formation in complex networks [14, 20].

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Among spectral invariants, the *Laplacian energy* of a graph of order n and size m, defined as

$$LE(G) = \sum_{i=1}^{n} |\lambda_i(L) - \overline{d}(G)|,$$

where $\overline{d}(G) = 2m/n$ is the average degree, was introduced by Gutman and Zhou [16] as a natural counterpart to Gutman's classical adjacency energy [15, 18, 24]. Laplacian equienergeticity itself remains an active topic: Popat and Tura recently characterized all graphs whose Laplacian energy equals that of their complement, obtaining several infinite families of such graphs [25]. For a vertex $v \in V(G)$, let $G^{(v)}$ denote the graph obtained by deleting v and all its incident edges. The *local Laplacian energy* at v is

$$\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)}),$$

quantifying the signed variation in Laplacian energy caused by deleting v. Unlike local adjacency energy, which is always non-negative [4, 13, 27], $\mathscr{LE}_G(v)$ is a signed variation; deleting a vertex can reduce or increase the global Laplacian energy. Summing over all vertices gives the *total local Laplacian energy*

$$e_L(G) = \sum_{v \in V(G)} \mathscr{LE}_G(v).$$

Two graphs G and H of the same order are called *local Laplacian equienergetic* if $e_L(G) = e_L(H)$, meaning that the sums of their vertex-deletion energy variations coincide. Beyond this first-order aggregate sum, the individual variations across the graph form the *local Laplacian energy profile* multiset

$$\mathcal{M}_L(G) = \{\mathscr{LE}_G(v) : v \in V(G)\}.$$

Two graphs are called *strongly local Laplacian equienergetic* if their local multisets coincide identically, $\mathcal{M}_L(G) = \mathcal{M}_L(H)$.

Vertex deletion is particularly natural for local spectral analysis because it isolates the energetic impact of a single node while evaluating the same invariant on a reduced-order graph. The study of energy changes under graph deletion operations originated in the adjacency setting. Day and So established foundational singular-value methods for edge deletion [8, 9], followed by refinements on edge perturbations [30]. For normalized Laplacian energy, Allem, Jacobs, and Trevisan obtained quantitative bounds for edge deletions [1]. At the node level, Arizmendi, Fernández-Hidalgo, and Juarez-Romero introduced the concept of vertex energy [4]. Espinal and Rada then investigated graph energy changes under vertex deletion [13], and Rada characterized the graphs for which the local vertex energies $E(G) - E(G - v)$, the adjacency analogue of the quantity studied here, attain the extremal sum $2E(G)$, introducing an adjusted local energy that sums exactly to $E(G)$ [27]. Rakshith and Das studied local graph energy bounds [28], and Dede and Popat investigated locally equienergetic graphs in the adjacency-energy setting [11].

The Laplacian counterpart has developed only recently. Because the trace of $L(G)$ depends directly on the number of edges, removing a vertex alters both the matrix dimension ($n \to n-1$) and the trace centering baseline ($\overline{d}(G) \to \overline{d}(G^{(v)}) = \frac{2(m-d_v)}{n-1}$), introducing dynamic spectral shifts absent in adjacency matrices. Moon and Park formally studied local Laplacian energy, derived bounds on vertex-deletion variations, and examined its relation to global Laplacian energy [23]. In a complementary direction, Dede and Popat investigated Laplacian energy changes under single edge deletions and classified small locally equienergetic pairs [12]. The present paper differs from these earlier studies in that it establishes the deletion-energy profile $\mathcal{M}_L(G)$ as an algebraic object in its own right, analyzes its higher-order moment hierarchy and variance, determines the sign of the local energy in terms of the degree of the deleted vertex, and investigates simultaneous equienergeticity with respect to both global and local Laplacian energy.

Motivation and Problem Statement. Understanding the spatial distribution of Laplacian energy across individual vertices provides deeper structural insights than aggregate scalar invariants. While classical equienergetic graphs have been widely cataloged, comparatively little is

known about graphs that simultaneously preserve global Laplacian energy and total local Laplacian energy. This dual perspective is essential for identifying vertices that exert disproportionate leverage on network cohesion and for characterizing graph perturbations that preserve spectral energy distributions.

Unlike adjacency vertex energy, local Laplacian energy is governed by a dynamic centering shift that can induce strictly negative values. A central problem is therefore to determine when the deletion of a vertex can increase the Laplacian energy, and how this depends on the degree of the deleted vertex. Furthermore, while the aggregate local energy e_L captures first-order variation, it leaves open the finer distribution of energy among vertices. This motivates examining the full profile $\mathcal{M}_L(G)$, its variance $\mathrm{Var}(\mathcal{M}_L(G))$, and its discriminatory capacity. Specifically, it is of fundamental interest to determine whether the local profile can break the classical Laplacian cospectrality barrier, distinguishing non-isomorphic graphs that global energy cannot separate, and to establish through finite-order classification how rigid simultaneous equienergetic classes are with respect to edge counts, degree sequences, and spectral polynomials.

Contributions. The contributions of this paper are the following.

(1) **An affine reduction of the profile to the deletion deck.** The invariants $LE(G)$, $e_L(G)$ and $\overline{D}_L(G)$ satisfy one affine relation, so equality of any two forces the third; and $\mathcal{M}_L(G) = \mathcal{M}_L(H)$ holds exactly when the decks agree after a rigid translation by $\tau = LE(H) - LE(G)$, with $\mathrm{Var}(\mathcal{M}_L) = \mathrm{Var}(\mathcal{D}_L)$ at the level of every central moment. Section 2.

(2) **The sign of the local energy.** Deleting a vertex never increases the adjacency energy, but it can increase the Laplacian energy, because the deletion also moves the centering baseline, by $\Delta\overline{d}_v = \frac{\overline{d}(G)-2d_v}{n-1}$. We prove that this shift alone decides when an increase is possible: if $d_v \le \overline{d}(G)/2$ then $\mathscr{LE}_G(v) \ge 0$, so only the deletion of a vertex heavier than half the average degree can increase the Laplacian energy. Complete graphs, cycles and conference graphs show that the condition is not sufficient. To our knowledge this is the first general sign criterion for the local Laplacian energy; the proof combines a variational description of LE with a majorisation of the vertex-deleted Laplacian spectrum obtained from Weyl's inequality. Section 3.

(3) **Closed formulas for classical families, with structural readings.** We evaluate $\mathscr{LE}_G(v)$ for complete multipartite graphs, strongly regular and conference graphs, cycles and paths, corona products and clique joins. The local energy is bounded above by $LE(G)$, with equality only at star centres, and reaches $-\frac{n}{2}+O(1)$ on $K_{r,r}$; for a strongly regular graph it measures only the first subconstituent; for a cycle it is positive and tends to $4/\pi$; for $K_n \circ K_1$ the profile has two atoms separating linearly in the order, so that the profile spread of a graph on N vertices ranges from 0 to order N; and in a windmill the hub and the blades carry opposite signs. Section 4.

(4) **Profile variance and cospectrality.** The profile differs from the uncentered deletion profile vertex by vertex by at most the baseline shift. We bound the discrepancy between $\mathrm{Var}(\mathcal{M}_L(G))$ and the variance of the uncentered deletion profile by the degree modulus $\kappa(G) = \sqrt{4\sigma_d^2 + \overline{d}(G)^2}$, show that this modulus is asymptotically sharp along stars, and deduce that a Laplacian cospectral pair is separated by $\mathcal{M}_L$ precisely when its two deletion decks differ. The smallest cospectral pairs that are not separated have nine vertices. Section 5.

(5) **An exhaustive census through order ten.** All connected graphs of order at most ten are classified under simultaneous (LE, e_L)-equivalence, yielding 17856 classes at order ten, and exactly 19 of the 24335 resulting pairs are certified, in exact arithmetic, to share the full profile. Section 5.2.

Organization. Section 2 develops the deletion-energy profile and reduces it to the deletion deck. Section 3 determines the sign of the local energy. Section 4 evaluates the profile on classical graph families. Section 5 treats the profile variance and the separation of Laplacian cospectral graphs, and closes with the finite-order census. Section 6 collects open problems.

Notation. Throughout, G is a finite simple graph of order $n = |V(G)|$ and size $m = |E(G)|$, with degrees d_v, minimum degree $\delta(G)$, maximum degree $\Delta(G)$ and average degree $\overline{d}(G) = 2m/n$. We write $\lambda_1 \le \cdots \le \lambda_n$ for the Laplacian eigenvalues of G and $\gamma_1 \le \cdots \le \gamma_{n-1}$ for those of $G^{(v)}$. Table 1 lists the symbols that recur across several sections; symbols introduced for a single family or a single proof are defined where they occur. Four conventions deserve explicit mention, because the same letters carry different meanings in different parts of the spectral literature. The letter δ always denotes the minimum degree and never the translation parameter of Theorem 2.4, which is written τ; $\Delta(G)$ always denotes the maximum degree, while the discriminant of a strongly regular parameter set is written Δ_{srg}; σ_d^2 is the degree variance, while Σ_2 denotes the part-size moment of a complete multipartite graph; and m always denotes the size of a graph, so that the order of a path is written ℓ.

TABLE 1. Symbols used across several sections.

Symbol	Meaning
$G^{(v)}$	graph obtained from G by deleting v and its incident edges
$LE(G)$	Laplacian energy $\sum_i \lvert\lambda_i - \overline{d}(G)\rvert$
$\mathscr{LE}_G(v)$	local Laplacian energy $LE(G) - LE(G^{(v)})$
$e_L(G)$	total local Laplacian energy $\sum_v \mathscr{LE}_G(v)$
$\mathcal{D}_L(G)$	vertex-deleted energy deck $\{LE(G^{(v)}) : v \in V(G)\}$
$\overline{D}_L(G)$	mean deletion-deck energy $\frac{1}{n}\sum_v LE(G^{(v)})$
$\mathcal{M}_L(G)$	local energy profile $\{\mathscr{LE}_G(v) : v \in V(G)\}$
$G \sim_L H$	$LE(G) = LE(H)$ and $e_L(G) = e_L(H)$
τ	translation parameter $LE(H) - LE(G)$
$\Delta\overline{d}_v$	baseline shift $\overline{d}(G^{(v)}) - \overline{d}(G) = \frac{\overline{d}(G) - 2d_v}{n-1}$
$\Delta\Lambda(G, v)$	uncentered deletion variation, Eq. (24)
$S_k(G)$	sum of the k largest Laplacian eigenvalues of G
σ_d^2	degree variance $\frac{1}{n}\sum_v (d_v - \overline{d}(G))^2$
$\kappa(G)$	degree modulus $\sqrt{4\sigma_d^2 + \overline{d}(G)^2}$

2. Deletion-Energy Profiles and Equienergeticity

For a connected graph $G = (V(G), E(G))$ of order n, let $G^{(v)}$ denote the subgraph obtained by deleting the vertex $v \in V(G)$. The Laplacian-energy profile of the vertex-deletion deck is the multiset

$$\mathcal{D}_L(G) := \{LE(G^{(v)}) : v \in V(G)\}, \tag{1}$$

with the mean vertex-deleted Laplacian energy defined by

$$\overline{D}_L(G) := \frac{1}{n} \sum_{v \in V(G)} LE(G^{(v)}). \tag{2}$$

Correspondingly, the local Laplacian energy profile of G is the multiset

$$\mathcal{M}_L(G) := \{\mathscr{LE}_G(v) : v \in V(G)\}, \tag{3}$$

where $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)})$. Two graphs G and H of the same order are called *local Laplacian equienergetic* if they share the same total local energy, i.e., $e_L(G) = e_L(H)$, where $e_L(G) = \sum_{v \in V(G)} \mathscr{LE}_G(v)$.

The global quantity $LE(G)$, the mean deleted energy $\overline{D}_L(G)$, and the total local energy $e_L(G)$ are coupled by a single affine relation, which already determines how the three invariants behave under comparison of two graphs.

Lemma 2.1 (Deletion-profile decomposition)**.** *Let G and H be graphs of the same order n.*

(1) *The invariants* LE, e_L, *and* $\overline{D}_L$ *satisfy*

$$e_L(G) = n\big(LE(G) - \overline{D}_L(G)\big), \tag{4}$$

and consequently

$$\frac{e_L(G) - e_L(H)}{n} = \big(LE(G) - LE(H)\big) - \big(\overline{D}_L(G) - \overline{D}_L(H)\big). \tag{5}$$

(2) G *and* H *are local Laplacian equienergetic* ($e_L(G) = e_L(H)$) *if and only if*

$$LE(G) - LE(H) = \overline{D}_L(G) - \overline{D}_L(H). \tag{6}$$

(3) *Among the three equalities*

$$\text{(i) } LE(G) = LE(H), \quad \text{(ii) } \overline{D}_L(G) = \overline{D}_L(H), \quad \text{(iii) } e_L(G) = e_L(H),$$

any two imply the third. In particular, for globally Laplacian equienergetic graphs, local equienergeticity is equivalent to mean deletion-deck equienergeticity.

Proof. Summing $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)})$ over all $v \in V(G)$ gives

$$e_L(G) = \sum_{v \in V(G)} \big(LE(G) - LE(G^{(v)})\big) = nLE(G) - n\overline{D}_L(G),$$

which is Eq. (4); applying the same relation to H and subtracting yields Eq. (5). Assertions (2) and (3) follow at once from Eq. (5). □

Remark 2.2 (No Laplacian analogue of the adjacency bound). For the adjacency energy the local vertex energies satisfy $\sum_v \big(E(G) - E(G-v)\big) \le 2E(G)$ for every graph G (see [27]). The corresponding inequality $e_L(G) \le 2LE(G)$ fails for the Laplacian energy. By Corollary 4.2(3), the stars satisfy $e_L(S_n) = 4n - 6$ and $2LE(S_n) = 4n - 8 + \frac{8}{n}$, so that

$$e_L(S_n) - 2LE(S_n) = 2 - \frac{8}{n} > 0 \qquad \text{for every } n \ge 5.$$

By Eq. (4), $e_L(G) \le 2LE(G)$ is equivalent to $\overline{D}_L(G) \ge \big(1 - \frac{2}{n}\big)LE(G)$, so the stars are graphs whose vertex-deleted subgraphs have, on average, markedly smaller Laplacian energy than the graph itself.

The results of this section have a single cumulative effect, stated at its close. Up to one additive constant, the profile $\mathcal{M}_L(G)$ carries exactly the information of the vertex-deleted energy deck $\mathcal{D}_L(G)$.

2.1. **Strong Local Equivalence and Deck Translation.** While equality of the first moment $e_L(G) = e_L(H)$ captures the aggregate local energy, it does not guarantee that the distribution of energy contributions among individual vertices is preserved. This motivates a finer multiset equivalence.

Definition 2.3 (Strong local Laplacian equienergeticity)**.** Two graphs G and H of the same order are called *strongly local Laplacian equienergetic* if their local Laplacian energy multisets coincide

$$\mathcal{M}_L(G) = \mathcal{M}_L(H). \tag{7}$$

For a graph G of order n, we associate with $\mathcal{D}_L(G)$ and $\mathcal{M}_L(G)$ the monic counting polynomials

$$Q_G(x) := \prod_{v \in V(G)} \big(x - LE(G^{(v)})\big), \qquad P_G(x) := \prod_{v \in V(G)} \big(x - \mathscr{LE}_G(v)\big). \tag{8}$$

Theorem 2.4 (Translation of the deletion-energy deck)**.** *Let* G *and* H *be graphs of order* n*, and set the* translation parameter $\tau := LE(H) - LE(G)$ *(the symbol* δ *is reserved for the minimum degree). Then*

$$P_G(x) = (-1)^n Q_G\big(LE(G) - x\big). \tag{9}$$

Moreover, the following assertions are equivalent.

(1) G *and* H *are strongly local Laplacian equienergetic* ($\mathcal{M}_L(G) = \mathcal{M}_L(H)$)*;*
(2) $\mathcal{D}_L(H) = \{a + \tau : a \in \mathcal{D}_L(G)\}$*;*
(3) $Q_H(x) = Q_G(x - \tau)$.

In particular, if $LE(G) = LE(H)$*, then* $\mathcal{M}_L(G) = \mathcal{M}_L(H)$ *if and only if* $\mathcal{D}_L(G) = \mathcal{D}_L(H)$.

Proof. Substituting $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)})$ into $P_G(x)$ gives

$$\begin{aligned} P_G(x) &= \prod_{v\in V(G)} \big(x - LE(G) + LE(G^{(v)})\big) \\ &= (-1)^n \prod_{v\in V(G)} \big(LE(G) - x - LE(G^{(v)})\big) = (-1)^n Q_G\big(LE(G) - x\big), \end{aligned}$$

which proves Eq. (9).

The multiset equality $\mathcal{M}_L(G) = \mathcal{M}_L(H)$ is equivalent to $P_G(x) = P_H(x)$. Using Eq. (9) for both G and H, this is equivalent to $Q_G(LE(G)-x) = Q_H(LE(H)-x)$. Setting $y = LE(H)-x$ yields $Q_H(y) = Q_G(y - \tau)$, which corresponds to a rigid translation of all roots of Q_G by $\tau = LE(H) - LE(G)$, completing the proof. □

Corollary 2.5 (Deletion-energy deck criterion)**.** *Let* G *and* H *be graphs of the same order* n *with identical vertex-deleted Laplacian energy decks,*

$$\mathcal{D}_L(G) = \mathcal{D}_L(H). \tag{10}$$

Then the following three conditions are equivalent.

(1) $LE(G) = LE(H)$.
(2) $e_L(G) = e_L(H)$.
(3) $\mathcal{M}_L(G) = \mathcal{M}_L(H)$.

Since the Laplacian spectrum of a graph determines its Laplacian energy, the hypothesis is implied by, and is strictly weaker than, the requirement that G *and* H *have identical vertex-deleted Laplacian spectral decks* $\{\mathrm{spec}(L(G^{(v)})) : v \in V(G)\} = \{\mathrm{spec}(L(H^{(u)})) : u \in V(H)\}$.

Proof. Equal decks have equal means, so (1) and (2) are equivalent by Lemma 2.1(3). By Theorem 2.4, (3) holds if and only if $\mathcal{D}_L(H)$ is the translate of $\mathcal{D}_L(G)$ by τ; since the two decks coincide, this happens exactly when $\tau = 0$, that is, when (1) holds. □

Let $p_k(G) := \sum_{v\in V(G)} \mathscr{LE}_G(v)^k$ denote the power sums of the profile. By Newton's identities a multiset of n real numbers is determined by its first n power sums [19, p. 23], [29, Proposition 7.7.1], so $\mathcal{M}_L(G) = \mathcal{M}_L(H)$ if and only if $p_k(G) = p_k(H)$ for $1 \le k \le n$. The *central* moments, on the other hand, transfer exactly to the deck.

Theorem 2.6 (Central moments of the profile)**.** *Let* G *be a graph of order* n *and put* $\mu_L(G) = e_L(G)/n$*. Then*

$$\sum_{v\in V(G)} \big(\mathscr{LE}_G(v) - \mu_L(G)\big)^k = (-1)^k \sum_{v\in V(G)} \big(LE(G^{(v)}) - \overline{D}_L(G)\big)^k \qquad (k \ge 1). \tag{11}$$

In particular $\mathrm{Var}(\mathcal{M}_L(G)) = \mathrm{Var}(\mathcal{D}_L(G))$.

Proof. By Lemma 2.1, $\mu_L(G) = LE(G) - \overline{D}_L(G)$, whence $\mathscr{LE}_G(v) - \mu_L(G) = -\big(LE(G^{(v)}) - \overline{D}_L(G)\big)$. Raising to the k-th power and summing over v gives Eq. (11). □

2.2. **Examples and Structural Limitations.** Subsection 2.1 determines what equality of profiles forces. The hypotheses involved are not vacuous. Indeed, LE and e_L are logically independent, e_L is strictly coarser than $\mathcal{M}_L$, and $\mathcal{M}_L$ does not require cospectrality.

Remark 2.7 (Logical independence of LE and e_L). Neither equality implies the other, and both failures occur at very small orders. Indeed $e_L(P_3) = e_L(K_3) = 6$ while $LE(P_3) = \frac{10}{3} \neq 4 = LE(K_3)$, and the paw $K_1 \vee (K_1 \sqcup K_2)$ and the diamond $K_4 - e$ both have $LE = 6$ while their total local energies are $\frac{32}{3}$ and $\frac{28}{3}$. Among connected graphs of orders $3, 4, 5, 6$ there are respectively $1, 0, 10, 15$ pairs with equal e_L and different LE, and $0, 3, 4, 152$ pairs with equal LE and different e_L.

Example 2.8 (The total local energy, the profile and the spectrum)**.** (a) The graphs of order 6 with `graph6` strings `EEvw` and `EE~w` have spectra $\{0,1,3,5,5,6\}$ and $\{0,2,3,5,6,6\}$, yet $LE = 12$ and $\overline{D}_L = \frac{43}{5}$ for both, hence $e_L = \frac{102}{5}$ for both. Their profiles differ,

$$\left\{\left(\tfrac{14}{5}\right)^{(2)}, \left(\tfrac{18}{5}\right)^{(3)}, 4\right\} \neq \left\{\left(\tfrac{16}{5}\right)^{(3)}, \left(\tfrac{18}{5}\right)^{(3)}\right\},$$

so equality of e_L does not imply equality of profiles.

(b) In the other direction, the graphs of order 7 with `graph6` strings `FCQVw` and `FCQvw`, of sizes 10 and 11, share $LE = \frac{92}{7}$, $e_L = 20$ and the profile $\left\{\left(\frac{8}{7}\right)^{(2)}, \left(\frac{22}{7}\right)^{(4)}, \frac{36}{7}\right\}$. Their Laplacian traces differ, so they are not cospectral. Strong local equienergeticity therefore does not require cospectrality.

What Section 2 establishes. Lemma 2.1 shows that LE, e_L and $\overline{D}_L$ form a rank-two system, so e_L adds nothing beyond LE and the mean of the deck; Theorem 2.4 identifies equality of profiles with a rigid translation of the deck; and Theorem 2.6 carries the identification through every central moment. Hence, *up to a single additive constant, the profile $\mathcal{M}_L(G)$ carries exactly the information of the vertex-deleted energy deck $\mathcal{D}_L(G)$*, and by Example 2.8 this deck is strictly finer both than its own mean and than the Laplacian spectrum. The rest of the paper studies the deck through its sign (Section 3), its evaluation on graph families (Section 4), and its dispersion and discriminating power (Section 5).

3. The Sign of the Local Laplacian Energy

Section 2 reduced the profile to the deletion deck. The first question the deck raises is one of sign. When can removing a vertex *increase* the Laplacian energy of a graph?

The question is not rhetorical. Unlike its adjacency counterpart, the Laplacian energy is measured from a baseline that the deletion itself moves. Deleting v removes eigenvalue mass, which tends to lower LE, but it also changes the average degree, which recentres every remaining eigenvalue and may raise LE. The two effects act in opposite directions, and $\mathscr{LE}_G(v) < 0$ occurs exactly when the second wins. Explicit families where it does are given in Section 4; balanced complete multipartite graphs $K_{r^{(t)}}$ with $r \geq 3$ are the simplest.

This section shows that the competition is decided, at the level of eligibility, by a single degree threshold. We begin with the baseline shift itself.

Lemma 3.1 (Centering shift)**.** *Let G be a connected graph of order $n \geq 3$ and size m with average degree $\overline{d}(G) = \frac{2m}{n}$. For any vertex $v \in V(G)$ with degree d_v, the shift in average degree $\Delta\overline{d}_v := \overline{d}(G^{(v)}) - \overline{d}(G)$ is identically*

$$\Delta\overline{d}_v = \frac{\overline{d}(G) - 2d_v}{n-1}.$$

Consequently, the baseline shift direction is determined by the critical threshold $d_v = \frac{\overline{d}(G)}{2}$.

1. *$\Delta\overline{d}_v > 0$ if and only if $d_v < \frac{\overline{d}(G)}{2}$ (peripheral / low-degree vertices);*
2. *$\Delta\overline{d}_v = 0$ if and only if $d_v = \frac{\overline{d}(G)}{2}$;*
3. *$\Delta\overline{d}_v < 0$ if and only if $d_v > \frac{\overline{d}(G)}{2}$ (dominant / high-degree hubs).*

Proof. Direct algebraic computation yields

$$\begin{aligned}\Delta\overline{d}_v &= \frac{2(m-d_v)}{n-1} - \frac{2m}{n} = \frac{2nm - 2nd_v - 2mn + 2m}{n(n-1)} \\ &= \frac{2m - 2nd_v}{n(n-1)} = \frac{\overline{d}(G) - 2d_v}{n-1}.\end{aligned}$$

The sign of $\Delta\overline{d}_v$ coincides strictly with the sign of the numerator $\overline{d}(G) - 2d_v$. □

We show that the baseline threshold $d_v = \overline{d}(G)/2$ isolated in Lemma 3.1 is not merely indicative but genuinely necessary, in that no vertex on the light side of the threshold can carry negative local energy. The proof rests on two elementary facts, a variational description of Laplacian energy and a majorisation of the vertex-deleted Laplacian spectrum.

Lemma 3.2 (Variational form of Laplacian energy)**.** *Let F be a graph of order $N \geq 1$ and size M, and for $0 \leq k \leq N$ let $S_k(F)$ denote the sum of the k largest Laplacian eigenvalues of F (with $S_0(F) = 0$). Then*

$$LE(F) = 2 \max_{0 \leq k \leq N} \big(S_k(F) - k\,\overline{d}(F)\big), \tag{12}$$

and the maximum is attained at $k^* = \#\{i : \lambda_i > \overline{d}(F)\}$.

Proof. Since $\sum_{i=1}^{N} \lambda_i = \operatorname{tr} L(F) = 2M = N\overline{d}(F)$, the centred deviations sum to zero, so the positive and negative parts of $\{\lambda_i - \overline{d}(F)\}$ have equal total mass and

$$LE(F) = \sum_{i=1}^{N} |\lambda_i - \overline{d}(F)| = 2 \sum_{i\,:\,\lambda_i > \overline{d}(F)} \big(\lambda_i - \overline{d}(F)\big) = 2\big(S_{k^*}(F) - k^*\overline{d}(F)\big).$$

Listing the eigenvalues in decreasing order, increasing k beyond k^* adds non-positive terms and decreasing k below k^* removes positive ones, so k^* attains the maximum in Eq. (12). □

Lemma 3.3 (Deletion majorisation)**.** *Let G be a graph of order $n \geq 2$ and let $v \in V(G)$. Then $S_k(G^{(v)}) \leq S_k(G)$ for every $1 \leq k \leq n-1$.*

Proof. Let G_v denote the spanning subgraph of G whose edges are exactly the d_v edges incident with v, and let $\widetilde{L}$ be the $n \times n$ matrix obtained from $L(G^{(v)})$ by inserting a zero row and a zero column at the position of v. Since the edge set of G is the disjoint union of $E(G^{(v)})$ and $E(G_v)$, and Laplacians are additive over edge sets,

$$L(G) = \widetilde{L} + L(G_v),$$

and $L(G_v)$ is positive semidefinite. Weyl's inequality therefore bounds the k-th largest eigenvalue of $L(G)$ below by that of $\widetilde{L}$, for every k. The spectrum of $\widetilde{L}$ is $\operatorname{spec}(L(G^{(v)})) \cup \{0\}$, and all Laplacian eigenvalues are non-negative, so the extra zero is a smallest entry and the k largest eigenvalues of $\widetilde{L}$ are the k largest of $L(G^{(v)})$ whenever $k \leq n-1$. Summing the first k of these inequalities gives $S_k(G^{(v)}) \leq S_k(G)$. □

Theorem 3.4 (A necessary condition for negative local energy)**.** *Let G be a graph of order $n \geq 2$ with average degree $\overline{d}(G)$, and let $v \in V(G)$. If*

$$d_v \leq \frac{\overline{d}(G)}{2},$$

then $\mathscr{LE}_G(v) \geq 0$. Equivalently, $\mathscr{LE}_G(v) < 0$ forces $d_v > \frac{\overline{d}(G)}{2}$, so only vertices on the heavy side of the baseline threshold of Lemma 3.1 can carry negative local Laplacian energy.

Proof. Write $a = \overline{d}(G) = \frac{2m}{n}$ and $a' = \overline{d}(G^{(v)}) = \frac{2(m-d_v)}{n-1}$. By Lemma 3.1, $\Delta\overline{d}_v = \frac{a-2d_v}{n-1}$, so the hypothesis $d_v \leq a/2$ is exactly the statement $a' \geq a$. Applying Lemma 3.2 to $G^{(v)}$, then $a' \geq a$, then Lemma 3.3, and finally Lemma 3.2 to G, we obtain

$$\begin{aligned} LE(G^{(v)}) = 2 \max_{0 \leq k \leq n-1} \big(S_k(G^{(v)}) - ka'\big) \;&\leq\; 2 \max_{0 \leq k \leq n-1} \big(S_k(G^{(v)}) - ka\big) \\ &\leq 2 \max_{0 \leq k \leq n-1} \big(S_k(G) - ka\big) \;\leq\; LE(G), \end{aligned}$$

the last step because the maximum in Eq. (12) for G also ranges over $k = n$. Hence $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)}) \geq 0$. □

The condition of Theorem 3.4 is necessary but far from sufficient, and the simplest witness needs nothing beyond the complete graph. Every vertex of K_n satisfies $d_v = n-1 > \frac{n-1}{2} = \overline{d}(K_n)/2$ and is therefore eligible, yet $LE(K_n) = 2(n-1)$ gives

$$\mathscr{LE}_{K_n}(v) = 2(n-1) - 2(n-2) = 2 > 0 \qquad \text{for every } n \geq 2.$$

Section 4 supplies two infinite families of the same kind, namely conference graphs, which are regular and satisfy the explicit positive lower bound (19), and cycles, which are 2-regular and have positive local energy for every $n \geq 3$ by Corollary 4.11. Within the eligible regime the sign is genuinely delicate. Every vertex of a balanced multipartite graph $K_{r^{(t)}}$ is eligible, yet by Corollary 4.2(4) its local energy is negative exactly when $r \geq 3$; for $K_{r,r}$ it even reaches $-\frac{n}{2} + O(1)$, by Corollary 4.3. We do not know of a purely degree-theoretic condition that is both necessary and sufficient, and we record this as Problem 6.1 in Section 6.

What Section 3 contributes. For the adjacency energy the sign question has a classical answer: $A(G^{(v)})$ is a principal submatrix of $A(G)$, the trace norm does not increase under compression, and so deleting a vertex never increases the energy; the local adjacency energy is always non-negative [4, 13, 27]. For the Laplacian energy this monotonicity fails, because $L(G^{(v)})$ is not a principal submatrix of $L(G)$ and the centering baseline moves with the deletion. Moon and Park [23] studied the local Laplacian energy and derived bounds on the deletion variation. Theorem 3.4 determines exactly how much of the adjacency picture survives: monotonicity holds at every vertex on the light side of the threshold $d_v = \overline{d}(G)/2$, for every graph, while complete graphs, conference graphs and cycles show that the threshold cannot be turned into a characterization. To our knowledge this is the first general sign criterion for the local Laplacian energy, and its proof needs nothing beyond Weyl's inequality and the variational form of Lemma 3.2.

Remark 3.5 (Where negative local energy occurs). Among the 119 611 146 pairs (G, v) with G connected of order at most ten, exactly 161 428 have $\mathscr{LE}_G(v) < 0$, all with $d_v > \overline{d}(G)/2$, and at every even order up to ten the most negative value is attained by $K_{r,r}$. Local Laplacian energy thus behaves as a cohesion penalty rather than a degree reward: it is non-negative at light vertices and turns negative only at heavy vertices of dense, nearly regular blocks such as $K_{r^{(t)}}$, whose removal costs much average degree and little connectivity.

4. Local Laplacian Energy of Classical Graph Families

Computing the deck of a specific graph requires the spectra of its vertex-deleted subgraphs. This section obtains them in closed form for five classical families. The formulas are a means rather than an end: each family answers a different question about the local energy.

- *Complete multipartite graphs* (Subsection 4.1) mark the extremes. The bound $\mathscr{LE}_G(v) \leq LE(G)$ is attained exactly at star centres, and balanced bipartite graphs reach $-\frac{n}{2} + O(1)$, the smallest value found among all connected graphs of even order up to ten.
- *Strongly regular graphs* (Subsection 4.2) admit an explicit vertex-deleted Laplacian spectrum, from which the local energy turns out to depend, within a parameter set, only on the first subconstituent. For conference graphs this gives positivity with an explicit lower bound, so the necessary condition of Section 3 is not sufficient.
- *Paths and cycles* (Subsection 4.3) give a closed form for $LE(P_\ell)$ free of integer parts, and show that deleting a vertex from a cycle always has positive cost, which tends to the universal constant $4/\pi$.
- *Corona products* (Subsection 4.4) show that the spread of the profile can grow linearly in the order; with the constant profiles of vertex-transitive graphs this fixes the range that Section 5 quantifies.
- *Clique joins and windmills* (Subsection 4.5) exhibit both signs in a single graph: the hub always has positive local energy, the blade vertices eventually negative, and the sign of e_L records which prevails.

4.1. **Complete Multipartite Graphs and Core Subfamilies.** Complete multipartite graphs have integral Laplacian spectra, so their local energies are rational and fully explicit. Their role here is to delimit the range of the local energy: the upper bound $\mathscr{LE}_G(v) \le LE(G)$ is attained exactly at star centres, and the balanced bipartite graphs give the most negative values found (Corollary 4.3).

For a positive integer partition vector $\mathbf{n} = (n_1, \dots, n_k)$ with $k \ge 2$, let $K_{\mathbf{n}} = K_{n_1,\dots,n_k}$ denote the complete k-partite graph of order $N = \sum_{i=1}^{k} n_i$ and size $m = \frac{1}{2}(N^2 - \Sigma_2)$, where $\Sigma_2 = \sum_{i=1}^{k} n_i^2$. The average degree is $\overline{d}(K_{\mathbf{n}}) = N - \frac{\Sigma_2}{N}$. Define the piecewise spectral functional

$$\Phi(\mathbf{n}) := N + \frac{(k-2)\Sigma_2}{N} + \sum_{i=1}^{k} (n_i - 1)\left|\frac{\Sigma_2}{N} - n_i\right|. \tag{13}$$

For any $i \in \{1, \dots, k\}$, let $\mathbf{n}^{[i]}$ denote the vector obtained by decrementing the i-th entry $n_i \to n_i - 1$ (omitting the entry if $n_i = 1$).

Proposition 4.1 (Complete multipartite graphs)**.** *Let $G = K_{n_1,\dots,n_k}$ have order $N \ge 2$. Then the global Laplacian energy is $LE(G) = \Phi(\mathbf{n})$. For any vertex v belonging to the i-th partite set V_i, the local Laplacian energy and total local energy are given by*

$$\mathscr{LE}_G(v) = \Phi(\mathbf{n}) - \Phi(\mathbf{n}^{[i]}), \qquad e_L(G) = \sum_{i=1}^{k} n_i\big(\Phi(\mathbf{n}) - \Phi(\mathbf{n}^{[i]})\big). \tag{14}$$

Proof. The Laplacian spectrum of G is $\operatorname{spec}(L(G)) = \{0^{(1)}, N^{(k-1)}, (N - n_1)^{(n_1-1)}, \dots, (N - n_k)^{(n_k-1)}\}$. Direct summation of the centered absolute deviations $|\lambda_j - \overline{d}(G)|$ yields $LE(G) = \Phi(\mathbf{n})$. Since deleting $v \in V_i$ produces the induced multipartite subgraph $G^{(v)} \cong K_{\mathbf{n}^{[i]}}$, the local energy formula follows from $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)})$. □

Evaluating Proposition 4.1 on canonical subfamilies gives explicit rational expressions.

Corollary 4.2 (Complete, star, and complete bipartite graphs)**.**

(1) ***Complete and empty graphs.*** *For the complete graph K_n ($n \ge 2$), $\mathscr{LE}_{K_n}(v) = 2$ for every vertex, yielding $e_L(K_n) = 2n$ and $\mathcal{M}_L(K_n) = \{2^{(n)}\}$.*

(2) ***Complete bipartite graphs.*** *Let $G = K_{p,q}$ with $p, q \ge 1$, and set*

$$\Lambda(p,q) = \frac{(p+q)^2 + |p-q|(2pq - p - q)}{p+q}.$$

Then every vertex x in the part of size p satisfies $\mathscr{LE}_{K_{p,q}}(x) = \Lambda(p,q) - \Lambda(p-1,q)$. If $1 \le p < q$, the total local energy is strictly positive,

$$e_L(K_{p,q}) = \frac{2p(2q-1)(q-p+1)}{p+q-1} > 0.$$

(3) ***Stars.*** *For $S_n \cong K_{1,n-1}$ with $n \ge 3$, the centre c and every leaf u satisfy*

$$LE(S_n) = \mathscr{LE}_{S_n}(c) = \frac{2(n^2 - 2n + 2)}{n},$$
$$\mathscr{LE}_{S_n}(u) = \frac{2(n-2)(n+1)}{n(n-1)},$$

yielding the constant-slope total $e_L(S_n) = 2(2n-3)$. In particular $e_L(S_3) = e_L(P_3) = 6$, in agreement with Remark 2.7.

(4) ***Balanced complete multipartite graphs.*** *For $t \ge 2$ parts of size $r \ge 1$ ($N = rt$), $LE(K_{r^{(t)}}) = 2r(t-1)$, and every vertex v satisfies*

$$\mathscr{LE}_{K_{r^{(t)}}}(v) = -\frac{2(t-1)(r^2 - 3r + 1)}{tr - 1}, \quad e_L(K_{r^{(t)}}) = tr\,\mathscr{LE}_{K_{r^{(t)}}}(v).$$

Proof. Parts (1), (2) and (4) follow by direct evaluation of Proposition 4.1 with $\mathbf{n} = (1^{(n)})$, $\mathbf{n} = (p, q)$ and $\mathbf{n} = (r^{(t)})$, respectively.

Part (3) is part (2) with $(p, q) = (1, n-1)$, which gives $\Lambda(1, n-1) = \frac{2(n^2-2n+2)}{n}$; deleting the centre leaves an edgeless graph, and deleting a leaf leaves S_{n-1}. □

The family also marks the two ends of the range of the local energy.

Corollary 4.3 (Range of the local energy)**.** *Let G be a connected graph of order $n \geq 2$ and let $v \in V(G)$. Then $\mathscr{LE}_G(v) \leq LE(G)$, with equality if and only if $G \cong K_{1,n-1}$ with centre v. On the other hand, for $G = K_{r,r}$ and $n = 2r$,*

$$\mathscr{LE}_{K_{r,r}}(v) = -\frac{n}{2} + \frac{5}{2} + \frac{1}{2(n-1)}, \tag{15}$$

which is negative for every $n \geq 6$.

Proof. Since $LE(G^{(v)}) \geq 0$, the inequality is clear. Equality forces every Laplacian eigenvalue of $G^{(v)}$ to equal $\overline{d}(G^{(v)})$, so that $L(G^{(v)})$ is a scalar matrix; its off-diagonal entries then vanish and $G^{(v)}$ is edgeless. A connected graph all of whose edges contain v is a star centred at v. Eq. (15) is Corollary 4.2(4) with $t = 2$, rewritten by means of $2(r^2 - 3r + 1) = (2r - 1)(r - \frac{5}{2}) - \frac{1}{2}$. □

The upper bound is attained exactly at star centres, where it equals $\frac{2(n^2-2n+2)}{n}$. At the other end every vertex of $K_{r,r}$ is eligible in the sense of Theorem 3.4, since $d_v = r = \overline{d}(G)$, and the balanced bipartite graphs realise the most negative local energy among all connected graphs of orders 4, 6, 8 and 10. Whether they do so for every even order is Problem 6.2.

4.2. **Regular, Vertex-Transitive, and Strongly Regular Graphs.** For strongly regular graphs we obtain the complete vertex-deleted Laplacian spectrum (Theorem 4.5). Its consequence is structural rather than computational: within a fixed parameter set the local energy is a function of the spectrum of the first subconstituent alone (Remark 4.9). For conference graphs it yields positivity at every vertex with an explicit lower bound (Corollary 4.8), which shows that the necessary condition of Theorem 3.4 is not sufficient.

When G is a connected k-regular graph of order n, its Laplacian and adjacency eigenvalues are linearly coupled by $\lambda_i(L) = k - \mu_{n-i+1}(A)$. Hence, the global Laplacian energy coincides with the classical adjacency energy $LE(G) = E(A(G))$. However, vertex deletion breaks regular symmetry, since in $G^{(v)}$ the k neighbours of v have degree $k-1$ while the remaining $n-1-k$ vertices retain degree k, resulting in an irregular baseline $\overline{d}(G^{(v)}) = k - \frac{k}{n-1}$.

Remark 4.4 (Vertex-transitive graphs). If G is vertex-transitive of order n, an automorphism mapping u to v induces an isomorphism $G^{(u)} \cong G^{(v)}$. Hence $\mathcal{D}_L(G)$ and $\mathcal{M}_L(G)$ are constant multisets, $\mathscr{LE}_G(v) = e_L(G)/n$ for every v, and $\mathrm{Var}(\mathcal{M}_L(G)) = 0$.

To compute explicit closed formulas for highly symmetric irregular subgraphs, we now turn to strongly regular graphs. A connected k-regular graph G is *strongly regular* with parameters (n, k, λ, μ), denoted $\mathrm{srg}(n, k, \lambda, \mu)$, if every adjacent pair shares λ common neighbors and every non-adjacent pair shares μ common neighbors. Its non-trivial adjacency eigenvalues are $r, s = \frac{(\lambda-\mu)\pm\sqrt{\Delta_{\mathrm{srg}}}}{2}$ with multiplicities $f, g = \frac{1}{2}[(n-1) \mp \frac{2k+(n-1)(\lambda-\mu)}{\sqrt{\Delta_{\mathrm{srg}}}}]$, where $\Delta_{\mathrm{srg}} := (\lambda-\mu)^2+4(k-\mu)$ is the discriminant of the strongly regular parameter set (not to be confused with the maximum degree $\Delta(G)$). The global Laplacian energy is $LE(G) = 2k + 2fr$.

Theorem 4.5 (Deletion spectrum of a strongly regular graph)**.** *Let $G = \mathrm{srg}(n, k, \lambda, \mu)$ be a connected strongly regular graph with $k \leq \min\{f, g\}$. For any $v \in V(G)$, let $\theta_0 \geq \theta_1 \geq \cdots \geq \theta_{k-1}$ be the adjacency eigenvalues of the first subconstituent $G[N(v)]$; since every vertex of $N(v)$ has exactly λ neighbours inside $N(v)$, the subgraph $G[N(v)]$ is λ-regular and therefore $\theta_0 = \lambda$ with eigenvector $\mathbf{1}_k$.*

The Laplacian spectrum of the vertex-deleted subgraph $G^{(v)}$ consists of the following eigenvalues.

(1) *The trivial eigenvalue* $\xi_0 = 0$ *(multiplicity* 1*);*
(2) *The quotient eigenvalue* $\xi_1 = k - 1 - \lambda + \mu$ *(multiplicity* 1*);*
(3) *The uncoupled bulk eigenvalues* $k - r$ *(multiplicity* $f - k \geq 0$*) and* $k - s$ *(multiplicity* $g - k \geq 0$*);*
(4) *The* $k - 1$ *conjugate eigenvalue pairs*

$$\xi_j^{\pm} = \frac{(2k - 1 - \lambda + \mu) \pm \sqrt{(1 + \lambda - \mu)^2 + 4(k - \lambda + \theta_j)}}{2}$$

for $j = 1, 2, \ldots, k - 1$.

With average degree $\overline{d}(G^{(v)}) = k - \frac{k}{n-1}$*, the vertex-deleted Laplacian energy is*

$$\begin{aligned} LE(G^{(v)}) = \overline{d}(G^{(v)}) + |\xi_1 - \overline{d}(G^{(v)})| + (f - k)\left|r - \tfrac{k}{n-1}\right| + (g - k)\left|\tfrac{k}{n-1} - s\right| \\ + \sum_{j=1}^{k-1} \left(|\xi_j^+ - \overline{d}(G^{(v)})| + |\xi_j^- - \overline{d}(G^{(v)})| \right). \end{aligned} \tag{16}$$

If G *is vertex-transitive, the local Laplacian energy at every vertex is* $\mathscr{LE}_G(v) = 2k + 2fr - LE(G^{(v)})$.

Proof. Partition $V(G^{(v)})$ into $V_1 = N(v)$ and $V_2 = V(G) \setminus (N(v) \cup \{v\})$. The equitable quotient matrix with respect to $\{V_1, V_2\}$ is $B_L = \begin{bmatrix} k - 1 - \lambda & -(k - 1 - \lambda) \\ -\mu & \mu \end{bmatrix}$, with eigenvalues $\xi_0 = 0$ and $\xi_1 = k - 1 - \lambda + \mu$.

Write $A_N = A(G[N(v)])$ and $A_2 = A(G[V_2])$, and let B be the $N(v) \times V_2$ block of $A(G)$. Two identities follow by counting common neighbours in $A(G)^2 = kI + \lambda A(G) + \mu(J - I - A(G))$. Two distinct vertices of $N(v)$ have the common neighbour v, their common neighbours in $N(v)$ and those in V_2; since every vertex of $N(v)$ has $k - 1 - \lambda$ neighbours in V_2, this gives

$$BB^T = (k - 1)I + (\lambda - \mu)A_N + (\mu - 1)(J - I) - A_N^2.$$

A vertex of V_2 and a vertex of $N(v)$ have no common neighbour v, so

$$A_2 B^T + B^T A_N = \lambda B^T + \mu(J - B^T).$$

Now let $w \perp \mathbf{1}_k$ be an eigenvector of A_N with eigenvalue θ_j. Since $Jw = 0$, the identities give

$$BB^T w = \beta_j w, \qquad A_2 B^T w = (\lambda - \mu - \theta_j)\, B^T w,$$

$$\beta_j := k - \mu + (\lambda - \mu)\theta_j - \theta_j^2 = \frac{\|B^T w\|^2}{\|w\|^2}.$$

In particular $\beta_j = (r - \theta_j)(\theta_j - s) \geq 0$, because $r + s = \lambda - \mu$, $rs = \mu - k$, and $s \leq \theta_j \leq r$ by Cauchy interlacing for the induced subgraph $G[N(v)]$. The vertices of $N(v)$ have degree $k - 1$ in $G^{(v)}$ and those of V_2 have degree k, so

$$L(G^{(v)}) \begin{bmatrix} w \\ 0 \end{bmatrix} = \begin{bmatrix} (k - 1 - \theta_j)\, w \\ -B^T w \end{bmatrix},$$

$$L(G^{(v)}) \begin{bmatrix} 0 \\ B^T w \end{bmatrix} = \begin{bmatrix} -\beta_j\, w \\ (k - \lambda + \mu + \theta_j)\, B^T w \end{bmatrix}.$$

Hence $\mathcal{W}_j = \operatorname{span}\{(w, 0)^T, (0, B^T w)^T\}$ is invariant under $L(G^{(v)})$, which acts on it as the 2×2 block

$$M_j = \begin{bmatrix} k - 1 - \theta_j & -\beta_j \\ -1 & k - \lambda + \mu + \theta_j \end{bmatrix}.$$

If $\beta_j > 0$ the subspace is two-dimensional and $L(G^{(v)})$ has the two roots of $\det(xI - M_j)$ on it. If $\beta_j = 0$, that is $\theta_j \in \{r, s\}$, then $B^T w = 0$, the vector $(w, 0)^T$ is an eigenvector with eigenvalue $k - 1 - \theta_j$, and the second root $k - \lambda + \mu + \theta_j$ equals $k - s$ or $k - r$, a bulk eigenvalue. In either case the roots of $\det(xI - M_j)$ are the pair $\xi_j^{\pm}$. When $k \leq \min\{f, g\}$ both bulk multiplicities $f - k$ and $g - k$ are non-negative, and the remaining $n - 1 - 2k$ orthogonal modes span the

uncoupled bulk spectrum with eigenvalues $k-r$ and $k-s$. The multiplicities always add up correctly, since

$$1+1+(f-k)+(g-k)+2(k-1)=f+g=n-1.$$

Summing the deviations from $\overline{d}(G^{(v)})$ establishes Eq. (16). □

Corollary 4.6 (Triangle-free strongly regular graphs)**.** *Let $G=\mathrm{srg}(n,k,0,\mu)$ be a triangle-free strongly regular graph. Then $G[N(v)]\cong\overline{K_k}$, so $\theta_j=0$ for all $j=1,\dots,k-1$. Consequently, $\xi_1=k-1+\mu$, $\xi^{\pm}=\frac{(2k-1+\mu)\pm\sqrt{(1-\mu)^2+4k}}{2}$ with multiplicity $k-1$, and*

$$LE(G^{(v)})=\overline{d}(G^{(v)})+|\xi_1-\overline{d}(G^{(v)})|+(f-k)\left|r-\tfrac{k}{n-1}\right|+(g-k)\left|\tfrac{k}{n-1}-s\right|+(k-1)\sqrt{(1-\mu)^2+4k}.$$

Proof. Setting $\lambda=0$ in Theorem 4.5 gives $\theta_j=0$ for $1\le j\le k-1$, hence $\xi_1=k-1+\mu$ and $\xi^{\pm}=\frac{(2k-1+\mu)\pm\sqrt{(1-\mu)^2+4k}}{2}$, each of multiplicity $k-1$. The midpoint of the pair is $\frac{2k-1+\mu}{2}$ and $\overline{d}(G^{(v)})=k-\frac{k}{n-1}$; since $\mu\ge 1$ and $n-1>k$ one has $\xi^-<\overline{d}(G^{(v)})<\xi^+$, so $|\xi^+-\overline{d}(G^{(v)})|+|\xi^--\overline{d}(G^{(v)})|=\xi^+-\xi^-=\sqrt{(1-\mu)^2+4k}$. Substituting into Eq. (16) gives the stated formula. □

Example 4.7 (The Petersen and Clebsch graphs)**.** (1) **Petersen graph** $\mathcal{P}=\mathrm{srg}(10,3,0,1)$**.** Here $\Delta_{\mathrm{srg}}=9$, $r=1$, $s=-2$, $f=5$, $g=4$ and $LE(\mathcal{P})=16$. As $\mu=1$, $\xi^{\pm}=3\pm\sqrt{3}$; with $\overline{d}(G^{(v)})=8/3$,

$$LE(\mathcal{P}^{(v)})=\tfrac{20}{3}+4\sqrt{3}\approx 13.594870,\quad \mathscr{LE}_{\mathcal{P}}(v)=\tfrac{28}{3}-4\sqrt{3}\approx 2.405130.$$

(2) **Clebsch graph** $\mathcal{C}=\mathrm{srg}(16,5,0,2)$**.** Here $\Delta_{\mathrm{srg}}=16$, $r=1$, $s=-3$, $f=10$, $g=5$, and $LE(\mathcal{C})=30$. Since $\mu=2$, we find $\xi_1=6$ and $\xi^{\pm}=\frac{11\pm\sqrt{21}}{2}$ (multiplicity 4). The bulk contributes $f-k=5$ copies of $k-r=4$ and $g-k=0$ copies of $k-s=8$. With $\overline{d}(G^{(v)})=k-\frac{k}{n-1}=5-\frac{5}{15}=\frac{14}{3}$,

$$LE(\mathcal{C}^{(v)})=\underbrace{\tfrac{14}{3}}_{\overline{d}(G^{(v)})}+\underbrace{\tfrac{4}{3}}_{|\xi_1-\overline{d}(G^{(v)})|}+\underbrace{\tfrac{10}{3}}_{\text{bulk}}+4\sqrt{21}=\frac{28}{3}+4\sqrt{21}\approx 27.663636,$$

so that $\mathscr{LE}_{\mathcal{C}}(v)=30-\frac{28}{3}-4\sqrt{21}=\frac{62}{3}-4\sqrt{21}\approx 2.336364$.

Corollary 4.8 (Conference and Paley graphs)**.** *Let G be a conference graph with parameters $(n,k,\lambda,\mu)=(4\mu+1,2\mu,\mu-1,\mu)$, and let $\theta_1,\dots,\theta_{(n-3)/2}$ be the non-principal adjacency eigenvalues of $G[N(v)]$. Then $f=g=k=2\mu$, the uncoupled bulk spectrum is empty ($f-k=g-k=0$), $\overline{d}(G^{(v)})=\frac{n-2}{2}$, and*

$$LE(G^{(v)})=\frac{n-1}{2}+\sum_{j=1}^{\frac{n-3}{2}}\sqrt{n+3+4\theta_j}\,, \tag{17}$$

so that

$$\mathscr{LE}_G(v)=\frac{(n-1)\sqrt{n}}{2}-\sum_{j=1}^{\frac{n-3}{2}}\sqrt{n+3+4\theta_j}\,. \tag{18}$$

Moreover the local energy of a conference graph is strictly positive at every vertex

$$\mathscr{LE}_G(v)\;\ge\;\frac{n-1}{2}\big(\sqrt{n}-\sqrt{n-3}\,\big)\;>\;0, \tag{19}$$

with equality in Eq. (19) if and only if the θ_j are all equal, which happens exactly for $n=5$.

Proof. Here $\lambda-\mu=-1$, so $\Delta_{\mathrm{srg}}=n$ and $f=g=\frac{n-1}{2}=k$, and both bulk multiplicities vanish; moreover $\overline{d}(G^{(v)})=\frac{n-2}{2}$ and $\xi_1=k=\frac{n-1}{2}$, so $|\xi_1-\overline{d}(G^{(v)})|=\frac12$. Since $1+\lambda-\mu=0$, $2k-1-\lambda+\mu=n-1$ and $4(k-\lambda+\theta_j)=n+3+4\theta_j$, Theorem 4.5 gives $\xi_j^{\pm}=\frac12\big((n-1)\pm\sqrt{n+3+4\theta_j}\big)$, a pair centred at $\overline{d}(G^{(v)})+\frac12$. Cauchy interlacing for the induced subgraph

$G[N(v)]$ gives $\theta_j \geq s = -\frac{1+\sqrt{n}}{2}$, hence $n+3+4\theta_j \geq (\sqrt{n}-1)^2 > 1$ because $n \geq 5$. Each pair therefore straddles $\overline{d}(G^{(v)})$ and contributes $\xi_j^+ - \xi_j^- = \sqrt{n+3+4\theta_j}$, which turns Eq. (16) into Eq. (17); subtracting it from $LE(G) = 2k + 2fr = \frac{n-1}{2}(1+\sqrt{n})$ gives Eq. (18). Finally $G[N(v)]$ is traceless with $\theta_0 = \lambda$, so $\sum_{j\geq 1}(n+3+4\theta_j) = \frac{n-3}{2}(n+3) - (n-5) = \frac{(n-1)^2}{2}$, and the Cauchy–Schwarz inequality bounds the sum in Eq. (18) by $\frac{(n-1)\sqrt{n-3}}{2}$, with equality exactly when all radicands coincide. □

For the Paley graph $P_{13} = \mathrm{srg}(13, 6, 2, 3)$ one has $(\theta_1, \ldots, \theta_5) = (1, 1, -1, -1, -2)$, and Eq. (18) gives $\mathscr{LE}_{P_{13}}(v) \approx 2.932405$, above the bound $6(\sqrt{13} - \sqrt{10}) \approx 2.659642$.

Remark 4.9 (What the local energy of a strongly regular graph measures). In Theorem 4.5 the bulk eigenvalues $k - r$ and $k - s$, their multiplicities, and $LE(G) = 2k + 2fr$ depend only on (n, k, λ, μ), whereas ξ_1 and the pairs $\xi_j^{\pm}$ depend on the spectrum of the first subconstituent $G[N(v)]$. Within a fixed parameter set, therefore, $\mathscr{LE}_G(v)$ is a function of $\mathrm{spec}(A(G[N(v)]))$ alone. Two strongly regular graphs with the same parameters are Laplacian cospectral, so LE cannot separate them; their profiles agree whenever corresponding first subconstituents are cospectral, and any difference between the profiles must come from the first subconstituents. Example 5.8 shows that a difference does occur for the 4×4 rook's graph and the Shrikhande graph, whose first subconstituents are $2K_3$ and C_6.

4.3. **Cycles and Paths.** For paths we give a closed form of $LE(P_\ell)$ that avoids integer-part functions (Proposition 4.10). For cycles this yields a qualitative statement: the local energy is positive for every $n \geq 3$ and converges to $4/\pi$ (Corollary 4.11), so the cost of deleting a vertex from a long cycle does not depend on its length.

For $\ell \geq 2$ put

$$\Pi_\ell := \begin{cases} 2\cot\left(\frac{\pi}{2\ell}\right), & \ell \text{ even}, \\ 2\csc\left(\frac{\pi}{2\ell}\right) - \frac{2}{\ell}, & \ell \text{ odd}. \end{cases}$$

(The index ℓ denotes the *order* of the path; throughout the rest of the paper m always denotes the size of a graph.) For $n \geq 3$, let Γ_n denote the closed form for $LE(C_n)$,

$$\Gamma_n := \begin{cases} 4\cot\left(\frac{\pi}{n}\right), & n \equiv 0 \pmod 4, \\ 4\csc\left(\frac{\pi}{n}\right), & n \equiv 2 \pmod 4, \\ 2\csc\left(\frac{\pi}{2n}\right), & n \text{ is odd}. \end{cases}$$

Proposition 4.10 (Cycles and paths)**.** *For the path P_ℓ of order $\ell \geq 2$, $LE(P_\ell) = \Pi_\ell$. For the cycle C_n ($n \geq 3$), every vertex v satisfies*

$$\mathscr{LE}_{C_n}(v) = \Gamma_n - \Pi_{n-1}, \qquad e_L(C_n) = n\big(\Gamma_n - \Pi_{n-1}\big),$$

yielding the constant local profile $\mathcal{M}_L(C_n) = \{(\Gamma_n - \Pi_{n-1})^{(n)}\}$.

Proof. The Laplacian eigenvalues of P_ℓ are $2 - 2\cos\theta_j$ with $\theta_j = j\pi/\ell$ $(0 \leq j \leq \ell - 1)$, and $\overline{d}(P_\ell) = 2 - \frac{2}{\ell}$, so $LE(P_\ell) = 2\sum_j |\cos\theta_j - \frac{1}{\ell}|$. Put $s = \lceil \ell/2 \rceil - 1$. For $j > s$ we have $\cos\theta_j \leq 0$, while $\cos\theta_s$ equals $\sin(\pi/\ell) \geq 2/\ell$ or $\sin(\pi/2\ell) > 1/\ell$ according as ℓ is even or odd, by Jordan's inequality $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. Hence $\cos\theta_j > 1/\ell$ exactly for $j \leq s$. Since $\sum_j \cos\theta_j = 1$, the deviations sum to zero and $LE(P_\ell) = 4\sum_{j=0}^{s}(\cos\theta_j - \frac{1}{\ell})$. Lagrange's identity $\sum_{j=0}^{s}\cos j\theta = \frac{1}{2} + \frac{\sin((s+\frac{1}{2})\theta)}{2\sin(\theta/2)}$ with $\theta = \pi/\ell$ then gives Π_ℓ in both cases. Since C_n is 2-regular, $LE(C_n) = \Gamma_n$, and deleting any vertex gives $C_n^{(v)} \cong P_{n-1}$. □

Although the cycle is eligible at every vertex in the sense of Theorem 3.4, since $2 > 1 = \overline{d}(C_n)/2$, its local energy is never negative, and it converges.

Corollary 4.11 (Cycles)**.** *For every $n \geq 3$ the local energy of C_n is strictly positive, and*

$$\mathscr{LE}_{C_n}(v) = \frac{4}{\pi} + O\left(\frac{1}{n}\right), \qquad e_L(C_n) = \frac{4n}{\pi} + O(1). \tag{20}$$

Proof. Since $\cot x$ and $\csc x$ are both $\frac{1}{x}+O(x)$ as $x\to 0$, each case of Γ_n equals $\frac{4n}{\pi}+O(\frac{1}{n})$ and each case of Π_ℓ equals $\frac{4\ell}{\pi}+O(\frac{1}{\ell})$; subtracting gives Eq. (20). For positivity, suppose first that n is odd. With $x=\frac{\pi}{2n}<y=\frac{\pi}{2n-2}$ we get $\Gamma_n=2\csc x>\frac{2}{x}>\frac{2}{y}>2\cot y=\Pi_{n-1}$. Suppose next that $n=2p$ with $p\geq 2$, and put $a=\frac{\pi}{2p}$ and $b=\frac{\pi}{4p-2}\leq\frac{\pi}{6}$. Then $\Gamma_n\geq 4\cot a>\frac{4}{a}-2a$, because $\cos a\geq 1-\frac{a^2}{2}$ and $\sin a<a$. Moreover $\csc b-\frac{1}{b}=\frac{b-\sin b}{b\sin b}\leq\frac{b^2}{6\sin b}\leq\frac{\pi b}{18}\leq\frac{\pi^2}{108}$, using $b-\sin b\leq\frac{b^3}{6}$ and $\sin b\geq\frac{3b}{\pi}$ on $[0,\frac{\pi}{6}]$. Hence

$$\Gamma_n-\Pi_{n-1}\;>\;\frac{4}{\pi}-\frac{\pi}{p}+\frac{2}{2p-1}-\frac{\pi^2}{54}=:\varphi(p).$$

Here $\varphi(2)>0.18$, and φ is increasing on $p\geq 2$ because $\frac{\pi}{p^2}>\frac{4}{(2p-1)^2}$ there, so $\varphi(p)>0$ for all $p\geq 2$. □

Numerically $\mathscr{LE}_{C_{1000}}(v)=1.270529$ against $4/\pi=1.273240$, so deleting a vertex from a long cycle costs a fixed amount, independent of the order. The minimum over all cycles is $\mathscr{LE}_{C_4}(v)=\frac{2}{3}$.

4.4. **Corona Products $G\circ K_1$ and Two-Level Local Profiles.** Coronas attach a pendant vertex to every vertex. Their profile has exactly two values, whose difference grows linearly in the order (Corollary 4.15). Together with the constant profiles of vertex-transitive graphs this shows that the profile spread of a connected graph on N vertices ranges from 0 to order N, and its variance from 0 to order N^2.

For a connected graph $G=(V(G),E(G))$ of order n and size m, the *corona product* $G\circ K_1$ joins each vertex $v_i\in V(G)$ to a unique pendant leaf $u_i\in U=\{u_1,\dots,u_n\}$. The resulting graph has order $2n$, size $m+n$, and average degree $\overline{d}(G\circ K_1)=\frac{\overline{d}(G)}{2}+1$.

Proposition 4.12 (Laplacian energy of $G\circ K_1$)**.** *Let* $\operatorname{spec}(L(G))=\{0=\lambda_1,\lambda_2,\dots,\lambda_n\}$. *The Laplacian spectrum of* $G\circ K_1$ *consists of* $0,2$, *and the* $2(n-1)$ *eigenvalues*

$$\lambda_i^{\pm}=\frac{\lambda_i+2\pm\sqrt{\lambda_i^2+4}}{2},\qquad i=2,3,\dots,n.$$

With $\overline{d}(G\circ K_1)=\frac{\overline{d}(G)}{2}+1$, *the global Laplacian energy is*

$$LE(G\circ K_1)=\overline{d}(G\circ K_1)+|2-\overline{d}(G\circ K_1)|+\sum_{i=2}^{n}\left(|\lambda_i^+-\overline{d}(G\circ K_1)|+|\lambda_i^--\overline{d}(G\circ K_1)|\right).$$

Proof. Order the vertices of $G\circ K_1$ as $V(G)$ followed by U. Then

$$L(G\circ K_1)=\begin{bmatrix}L(G)+I_n & -I_n\\ -I_n & I_n\end{bmatrix}.$$

Let $\mathbf{w}$ be an eigenvector of $L(G)$ with eigenvalue λ_i. For $\alpha,\beta\in\mathbb{R}$ the vector $(\alpha\mathbf{w},\beta\mathbf{w})^T$ is an eigenvector of $L(G\circ K_1)$ with eigenvalue x precisely when

$$\begin{bmatrix}\lambda_i+1 & -1\\ -1 & 1\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}=x\begin{bmatrix}\alpha\\ \beta\end{bmatrix},$$

so each λ_i contributes the two roots of $x^2-(\lambda_i+2)x+\lambda_i=0$, namely $\lambda_i^{\pm}=\frac{\lambda_i+2\pm\sqrt{\lambda_i^2+4}}{2}$. Since the n eigenvectors of $L(G)$ span $\mathbb{R}^n$, these $2n$ values exhaust the spectrum. For $\lambda_1=0$ they reduce to $\lambda_1^-=0$ and $\lambda_1^+=2$, which gives the stated list. The energy formula follows by summing $|x-\overline{d}(G\circ K_1)|$ with $\overline{d}(G\circ K_1)=\frac{2(m+n)}{2n}=\frac{\overline{d}(G)}{2}+1$. □

Theorem 4.13 (Local profile of $K_n\circ K_1$)**.** *For* $n\geq 3$, *the complete graph corona* $K_n\circ K_1$ *has average degree* $\overline{d}(K_n\circ K_1)=\frac{n+1}{2}$, *and its global Laplacian energy is* $LE(K_n\circ K_1)=(n-1)(1+\sqrt{n^2+4})$. *The local Laplacian energy profile partitions into two distinct constant values*

$$\mathcal{M}_L(K_n\circ K_1)=\left\{\mathscr{LE}(v)^{(n)},\mathscr{LE}(u)^{(n)}\right\},$$

where for every base vertex $v \in V(K_n)$ and every pendant leaf $u \in U$

$$\mathscr{LE}_{K_n\circ K_1}(v) = (n-1)\big(1+\sqrt{n^2+4}\big) - \left[\frac{2n(n-1)}{2n-1} + \left|2 - \frac{n(n-1)}{2n-1}\right| + (n-2)\sqrt{(n-1)^2+4}\right],$$

$$\mathscr{LE}_{K_n\circ K_1}(u) = \frac{n^2-4n+3}{2n-1} + \sqrt{n^2+4} - \sqrt{n^2-4n+8}.$$

Proof. Deleting a base vertex $v \in V(K_n)$ produces $(K_n\circ K_1)^{(v)} \cong (K_{n-1}\circ K_1)\sqcup K_1$. Its Laplacian spectrum consists of $0^{(2)}, 2^{(1)}$, and the $2(n-2)$ eigenvalues $\frac{n+1\pm\sqrt{(n-1)^2+4}}{2}$. Evaluating the sum of deviations with average degree $\overline{d}'_v = \frac{n(n-1)}{2n-1}$ yields $\mathscr{LE}(v)$.

Deleting a pendant leaf u leaves the graph H obtained from K_n by attaching a pendant vertex at $n-1$ of its vertices. It has $\overline{d}(H) = \frac{(n+2)(n-1)}{2n-1}$ and $\varphi_{L(H)}(x) = x\left(x^2-(n+2)x+n\right)^{n-2}\big(x^2-(n+2)x+2n-1\big)$. Both quadratics have their roots on either side of $\overline{d}(H)$, so $LE(H) = \overline{d}(H)+(n-2)\sqrt{n^2+4}+\sqrt{n^2-4n+8}$, and subtracting this from $LE(K_n\circ K_1)$ yields $\mathscr{LE}(u)$. □

Example 4.14 (Local profiles of $K_3\circ K_1$ and $K_4\circ K_1$)**.** Theorem 4.13 gives the exact two-level profiles

	$LE(G)$	$\mathscr{LE}(v)$ (core)	$\mathscr{LE}(u)$ (pendant)
$K_3\circ K_1$	$2(1+\sqrt{13})$	$2\sqrt{13}-2\sqrt{2}-\frac{6}{5}$	$\sqrt{13}-\sqrt{5}$
$K_4\circ K_1$	$3+6\sqrt{5}$	$6\sqrt{5}-2\sqrt{13}-\frac{5}{7}$	$\frac{3}{7}+2\sqrt{5}-2\sqrt{2}$

that is $\{3.182675, 1.369483^{(3)}\}$ and $\{5.491020, 2.072280^{(4)}\}$ numerically. All local energies are strictly positive and core vertices dominate pendant ones.

The two atoms of the profile separate linearly in the order.

Corollary 4.15 (Profile spread of $K_n\circ K_1$)**.** *As $n\to\infty$,*

$$\mathscr{LE}_{K_n\circ K_1}(v) = \tfrac{3}{2}n - \tfrac{1}{4} + O\big(\tfrac{1}{n}\big), \qquad \mathscr{LE}_{K_n\circ K_1}(u) = \tfrac{1}{2}n + \tfrac{1}{4} + O\big(\tfrac{1}{n}\big).$$

Consequently, on $N = 2n$ vertices,

$$\max\mathcal{M}_L - \min\mathcal{M}_L = \frac{N-1}{2} + O\left(\frac{1}{N}\right), \qquad \mathrm{Var}(\mathcal{M}_L) = \frac{(N-1)^2}{16} + O(1). \tag{21}$$

Proof. Insert $\sqrt{n^2+4} = n+O(\frac{1}{n})$, $\sqrt{(n-1)^2+4} = n-1+O(\frac{1}{n})$, $\sqrt{n^2-4n+8} = n-2+O(\frac{1}{n})$ and $\frac{2n(n-1)}{2n-1} = n-\frac{1}{2}+O(\frac{1}{n})$ into Theorem 4.13. Each atom has multiplicity n, so the variance is the square of half the spread. □

In contrast with Remark 4.4, where the spread vanishes, the corona family shows that on N vertices the profile spread of a connected graph ranges from 0 to order N, and its variance from 0 to order N^2. Section 5 quantifies this dispersion. Corona products are not, however, extremal for the bound of Theorem 5.2, because for $K_n\circ K_1$ the standard deviations of $\mathcal{M}_L$ and of the uncentered profile are asymptotically equal, and the extremal family there is the stars (Remark 5.3).

4.5. **Clique Joins and Generalized Windmill Graphs.** Clique joins contain the generalized windmills, in which a single hub is joined to many cliques. In these graphs both signs of the local energy occur together: the hub always has positive local energy, whereas for blades of size at least three the blade vertices eventually have negative local energy (Corollary 4.18). The sign of the total e_L is decided by the competition between the two.

For $a \geq 1$ and a tuple $\mathbf{b} = (b_1, \ldots, b_r)$ ($b_i \geq 1$), the *clique join* is defined as $J(a;\mathbf{b}) := K_a \vee (K_{b_1}\sqcup\cdots\sqcup K_{b_r})$. Let $B = \sum_{i=1}^r b_i$ and $N = a+B$, and write the average degree as

$\overline{d}(a;\mathbf{b}) = \frac{a(a-1)+\sum_{i=1}^{r} b_i(b_i-1)+2aB}{N}$ (the symbol δ is reserved for the minimum degree). Define

$$\Omega(a;\mathbf{b}) := \overline{d}(a;\mathbf{b}) + a\big|N - \overline{d}(a;\mathbf{b})\big| + (r-1)\big|a - \overline{d}(a;\mathbf{b})\big| + \sum_{i=1}^{r}(b_i-1)\big|a+b_i - \overline{d}(a;\mathbf{b})\big|.$$

Proposition 4.16 (Clique joins)**.** *For any clique join $G = J(a;\mathbf{b})$, $LE(G) = \Omega(a;\mathbf{b})$. For a central vertex $x \in V(K_a)$ and a blade vertex $y \in V(K_{b_i})$*

$$\mathscr{LE}_G(x) = \Omega(a;\mathbf{b}) - \Omega(a-1;\mathbf{b}), \qquad \mathscr{LE}_G(y) = \Omega(a;\mathbf{b}) - \Omega(a;\mathbf{b}^{[i]}).$$

Proof. The Laplacian spectrum of $J(a;\mathbf{b})$ is

$$\{0^{(1)}, N^{(a)}, a^{(r-1)}, (a+b_1)^{(b_1-1)}, \dots, (a+b_r)^{(b_r-1)}\}.$$

Summing deviations from $\overline{d}(a;\mathbf{b})$ yields $LE(J(a;\mathbf{b})) = \Omega(a;\mathbf{b})$. Deleting a central vertex reduces $a \to a-1$, while deleting a blade vertex replaces $\mathbf{b} \to \mathbf{b}^{[i]}$. □

The *generalized windmill graph* with r blades of clique order $b+1$ is $W_{r,b} := K_1 \vee rK_b = J(1;(b,\dots,b))$.

Corollary 4.17 (Generalized windmills and friendship graphs)**.** *Let $b \geq 2$ and $r \geq 1$. Then $LE(W_{r,b}) = \frac{2(b^2r^2+br-r+1)}{br+1}$. For $r = 1$, $e_L(W_{1,b}) = 2(b+1)$. For $r \geq 2$*

$$e_L(W_{r,b}) = 2\big(-b^2r + 2b^2 + 3br - 4b + 1\big).$$

In particular, for the friendship graph $F_r = W_{r,2} = K_1 \vee rK_2$, the total local energy $e_L(F_r) = 2(2r+1) = 2|V(F_r)|$ is positive. Moreover $e_L(W_{r,b}) < 0$ holds if and only if

$$b \geq 4 \quad \text{and} \quad r > \frac{2b^2-4b+1}{b(b-3)}, \tag{22}$$

so that $b = 3$ never produces a negative total, while $b = 4$ requires $r \geq 5$, $b = 5$ requires $r \geq 4$, and $b \geq 6$ requires $r \geq 3$.

Proof. Apply Proposition 4.16 with $a = 1$ and $\mathbf{b} = (b,\dots,b)$, so that $N = br+1$ and $\overline{d}(1;\mathbf{b}) = \frac{b(b-1)r+2br}{br+1} = \frac{br(b+1)}{br+1}$. Substituting into Ω and simplifying gives $LE(W_{r,b}) = \frac{2(b^2r^2+br-r+1)}{br+1}$. Deleting the hub leaves rK_b, and deleting a blade vertex replaces one block K_b by K_{b-1}; summing the resulting $1+br$ local energies yields the stated expression for $e_L(W_{r,b})$ (for $r = 1$ the graph is K_{b+1} and Corollary 4.2(1) applies). Finally, $-b^2r + 2b^2 + 3br - 4b + 1 < 0$ rearranges to $b(b-3)r > 2b^2 - 4b + 1$, which for $b = 3$ reads $0 > 7$ and is impossible, and for $b \geq 4$ is Eq. (22). □

The sign of $e_L(W_{r,b})$ hides two opposite contributions, and they can be separated.

Corollary 4.18 (Hub and blades of a windmill)**.** *Let $b \geq 2$ and $r \geq 2$. The hub c and every blade vertex u of $W_{r,b}$ satisfy*

$$\begin{aligned} \mathscr{LE}(c) &= \frac{2(br^2+1)}{br+1}, \\ \mathscr{LE}(u) &= \frac{2\big(-(b^2-3b+1)\,r^2 + (2b^2-5b+4)\,r + 2(b-2)\big)}{r(br+1)}. \end{aligned} \tag{23}$$

In particular $\mathscr{LE}(c) = 2r + O(1)$ is positive, while $\mathscr{LE}(u) \to -\frac{2(b^2-3b+1)}{b}$ as $r \to \infty$, a limit that is negative for every $b \geq 3$ and equals $+1$ for the friendship graphs.

Proof. Deleting the hub leaves rK_b, whose spectrum is $\{0^{(r)}, b^{(r(b-1))}\}$ with average degree $b-1$, so $LE(rK_b) = 2r(b-1)$; subtracting from $LE(W_{r,b})$ gives $\mathscr{LE}(c)$. Since $e_L(W_{r,b}) = \mathscr{LE}(c) + br\,\mathscr{LE}(u)$, the blade value is $\big(e_L(W_{r,b}) - \mathscr{LE}(c)\big)/(br)$, which simplifies to Eq. (23). The limits follow on dividing by r^2. □

Both kinds of vertex are eligible in the sense of Theorem 3.4, since the hub has degree br and each blade vertex degree b, while $\overline{d}(W_{r,b}) < b+1$. Yet they behave oppositely. The hub always lowers the energy when deleted and does so by an amount growing linearly in r, whereas for $b \geq 3$ each blade vertex eventually raises it. The condition (22) records which of the two effects prevails, one hub against br blade vertices.

5. Profile Variance and the Discriminating Power of the Profile

Section 4 showed that the profile ranges from constant, for vertex-transitive graphs, to a spread of order N, for corona products. This section quantifies that dispersion and asks what the profile can see that LE cannot.

Subsection 5.1 compares the profile with the uncentered deletion profile $\Delta\Lambda$, in which the deleted eigenvalues are measured from the fixed baseline $\overline{d}(G)$. The two differ vertex by vertex by at most the baseline shift (Proposition 5.1), and in standard deviation by at most the degree modulus $\kappa(G) = \sqrt{4\sigma_d^2 + \overline{d}(G)^2}$ (Theorem 5.2). The modulus is asymptotically sharp along stars, regularity alone does not make the two variances equal, and a minimum degree below half the average forces a positive variance as soon as some local energy is negative.

Subsection 5.2 turns to Laplacian cospectral graphs, where LE carries no information. Cospectral graphs have the same profile exactly when their vertex-deleted energy decks agree. The profile separates the smallest cospectral pair, and the 4×4 rook's graph from the Shrikhande graph, where the variance cannot; but it is not a complete invariant, since the smallest cospectral graphs with identical profiles have nine vertices. An exhaustive census through order ten measures how selective the profile is, and every coincidence it records is certified exactly.

5.1. Profile Variance and Degree Statistics.

The aim of this subsection is to show that the dispersion of the profile is governed by the degree statistics of G, up to the dispersion of the uncentered deletion profile, and that this control is sharp. For a connected graph G of order $n \geq 3$ the profile variance is $\mathrm{Var}(\mathcal{M}_L(G)) = \frac{1}{n}p_2(G) - \mu_L(G)^2$, and by Theorem 2.6 it equals $\mathrm{Var}(\mathcal{D}_L(G))$.

To decouple the intrinsic spectral fluctuation from the movement of the baseline, we compare the deletion against the *fixed* baseline $\overline{d}(G)$. Define the *uncentered deletion variation*

$$\Delta\Lambda(G,v) := LE(G) - \sum_{j=1}^{n-1} \left|\gamma_j(L(G^{(v)})) - \overline{d}(G)\right|, \tag{24}$$

which differs from $\mathscr{LE}_G(v)$ only in that the deleted eigenvalues are measured from $\overline{d}(G)$ rather than from $\overline{d}(G^{(v)})$. The discrepancy between the two is controlled by the baseline shift of Lemma 3.1.

Proposition 5.1 (Effect of the centering shift)**.** *Let G be a connected graph of order $n \geq 3$, and let $v \in V(G)$ have degree d_v. Then*

$$|\mathscr{LE}_G(v) - \Delta\Lambda(G,v)| \leq |\overline{d}(G) - 2d_v|. \tag{25}$$

Proof. Write $a = \overline{d}(G)$ and $a' = \overline{d}(G^{(v)})$, and let $\gamma_1, \ldots, \gamma_{n-1}$ be the Laplacian eigenvalues of $G^{(v)}$. For every real x the reverse triangle inequality gives $\big||x-a'| - |x-a|\big| \leq |a'-a|$. Taking $x = \gamma_j$ and summing over j,

$$\left|LE(G^{(v)}) - \sum_{j=1}^{n-1}|\gamma_j - a|\right| = \left|\sum_{j=1}^{n-1}\big(|\gamma_j - a'| - |\gamma_j - a|\big)\right| \leq (n-1)\,|a'-a|.$$

By Lemma 3.1, $(n-1)|a'-a| = (n-1)|\Delta\overline{d}_v| = |\overline{d}(G) - 2d_v|$. Finally, by Eq. (24) and the definition of $\mathscr{LE}_G(v)$,

$$\mathscr{LE}_G(v) - \Delta\Lambda(G,v) = \sum_{j=1}^{n-1}|\gamma_j - a| - LE(G^{(v)}),$$

which is the quantity just bounded; this proves Eq. (25). □

Theorem 5.2 (Variance bound via degree statistics)**.** *Let G be a connected graph of order $n \geq 3$ with average degree $\overline{d}(G) = \frac{2m}{n}$ and topological degree variance $\sigma_d^2 := \frac{1}{n}\sum_{v\in V(G)}(d_v - \overline{d}(G))^2$. Put*

$$\kappa(G) := \sqrt{4\sigma_d^2 + \overline{d}(G)^2} \;=\; \Big(\frac{1}{n}\sum_{v\in V(G)}\big(\overline{d}(G) - 2d_v\big)^2\Big)^{1/2}. \tag{26}$$

Then the local Laplacian energy variance satisfies

$$\Big|\sqrt{\mathrm{Var}(\mathcal{M}_L(G))} - \sqrt{\mathrm{Var}(\Delta\Lambda(G))}\Big| \leq \kappa(G), \tag{27}$$

where $\mathrm{Var}(\Delta\Lambda(G))$ is the variance of the uncentered deletion profile $\{\Delta\Lambda(G,v)\}_v$. In particular

$$\mathrm{Var}(\mathcal{M}_L(G)) \leq 2\,\mathrm{Var}(\Delta\Lambda(G)) + 2\kappa(G)^2 = 2\,\mathrm{Var}(\Delta\Lambda(G)) + 8\sigma_d^2 + 2\overline{d}(G)^2. \tag{28}$$

Proof. For each $v \in V(G)$, define the baseline error $\varepsilon_v := \mathscr{LE}_G(v) - \Delta\Lambda(G,v)$. By Proposition 5.1, $|\varepsilon_v| \leq |\overline{d}(G) - 2d_v|$. We express the three profiles in terms of centred variables,

$$X_v = \mathscr{LE}_G(v) - \mu_L(G), \qquad Y_v = \Delta\Lambda(G,v) - \overline{\Delta\Lambda}, \qquad Z_v = \varepsilon_v - \bar{\varepsilon},$$

in the inner product space $\mathbb{R}^n$ endowed with $\langle \mathbf{u}, \mathbf{w}\rangle = \frac{1}{n}\sum_{i=1}^n u_i w_i$. The associated norms are the standard deviations $\|X\| = \sqrt{\mathrm{Var}(\mathcal{M}_L(G))}$, $\|Y\| = \sqrt{\mathrm{Var}(\Delta\Lambda(G))}$ and $\|Z\| = \sqrt{\mathrm{Var}(\varepsilon)}$, and $X = Y + Z$. The reverse triangle inequality gives $\big|\|X\| - \|Y\|\big| \leq \|Z\|$.

It remains to bound $\|Z\|$. Since the variance of a real vector never exceeds its second moment,

$$\mathrm{Var}(\varepsilon) \leq \frac{1}{n}\sum_{v\in V(G)}\varepsilon_v^2 \leq \frac{1}{n}\sum_{v\in V(G)}\big(\overline{d}(G) - 2d_v\big)^2.$$

Writing $2d_v - \overline{d}(G) = 2\big(d_v - \overline{d}(G)\big) + \overline{d}(G)$ and using $\sum_v\big(d_v - \overline{d}(G)\big) = 0$, the cross term vanishes and

$$\frac{1}{n}\sum_{v\in V(G)}\big(\overline{d}(G) - 2d_v\big)^2 = 4\sigma_d^2 + \overline{d}(G)^2 = \kappa(G)^2, \tag{29}$$

which is Eq. (26). Hence $\|Z\| \leq \kappa(G)$, establishing Eq. (27). Applying $(y+z)^2 \leq 2y^2 + 2z^2$ to $X_v = Y_v + Z_v$ and averaging gives Eq. (28). □

Remark 5.3 (Sharpness of the modulus). Neither part of $\kappa(G)$ can be dropped. The term $\overline{d}(G)^2$ is needed even on regular graphs, where $\sigma_d = 0$; Example 5.5 below exhibits a 4-regular graph with $|\mathrm{sd}(\mathcal{M}_L) - \mathrm{sd}(\Delta\Lambda)| = \frac{8\sqrt{3}}{21} > 0$. And the modulus as a whole is asymptotically attained. Along the stars, $\sigma_d^2 = \frac{(n-1)(n-2)}{n} \sim n$ and $\overline{d}(S_n) \to 2$, so $\kappa(S_n) \sim 2\sqrt{n}$; the profile consists of the value $LE(S_n) \sim 2n$ at the centre and $n-1$ values tending to 2 at the leaves, so $\mathrm{sd}(\mathcal{M}_L(S_n)) \sim 2\sqrt{n}$, while $\mathrm{sd}(\Delta\Lambda(S_n)) \to 0$. The ratio of the two sides of Eq. (27) thus tends to 1; it equals 0.650 at $n = 7$, 0.875 at $n = 20$ and 0.950 at $n = 50$. Among all 994 connected graphs of order at most seven its maximum, 0.650, is attained by S_7, although its median is only 0.065. The bound is thus asymptotically sharp but typically slack.

Corollary 5.4 (Regular graphs)**.** *Let G be a k-regular graph of order $n \geq 3$. Then $\sigma_d^2 = 0$ and $\kappa(G) = k$, so*

$$\Big|\sqrt{\mathrm{Var}(\mathcal{M}_L(G))} - \sqrt{\mathrm{Var}(\Delta\Lambda(G))}\Big| \leq k,$$
$$\mathrm{Var}(\mathcal{M}_L(G)) \leq 2\,\mathrm{Var}(\Delta\Lambda(G)) + 2k^2.$$

If in addition G is vertex-transitive, then both variances vanish and G is Laplacian deck-homogeneous. *Regularity alone does not force $\mathrm{Var}(\mathcal{M}_L(G)) = \mathrm{Var}(\Delta\Lambda(G))$, nor deck-homogeneity.*

Proof. The displayed inequalities are Theorem 5.2 with $\sigma_d = 0$ and $\overline{d}(G) = k$. If G is vertex-transitive, Remark 4.4 gives $\mathrm{Var}(\mathcal{M}_L(G)) = 0$, and the same automorphism argument applied to the uncentered sums $\sum_j |\gamma_j(L(G^{(v)})) - k|$ gives $\mathrm{Var}(\Delta\Lambda(G)) = 0$. The final assertion is Example 5.5. □

Example 5.5 (A regular graph that is not deck-homogeneous)**.** Let G be the connected 4-regular graph of order 7 and size 14 with `graph6` string `FFzn_`. Its local profile and uncentered deletion profile are

$$\mathcal{M}_L(G) = \left\{\tfrac{2}{3}^{(3)}, \tfrac{4}{3}^{(4)}\right\}, \qquad \Delta\Lambda(G) = \left\{0^{(4)}, 2^{(3)}\right\},$$

so that $\mathrm{Var}(\mathcal{M}_L(G)) = \frac{16}{147} \neq \frac{48}{49} = \mathrm{Var}(\Delta\Lambda(G))$, and the standard deviations differ by $\frac{4\sqrt{3}}{7} - \frac{4\sqrt{3}}{21} = \frac{8\sqrt{3}}{21}$. The graph is regular but not vertex-transitive. For a k-regular graph the deleted baseline is $\bar{d}(G^{(v)}) = k - \frac{k}{n-1} \neq k$, so the error ε_v is in general non-zero, and without an automorphism moving v to every other vertex it need not be constant in v.

Theorem 3.4 gives a second, purely combinatorial, route to a strictly positive profile variance.

Corollary 5.6 (Profile spread from the degree baseline)**.** *Let G be a connected graph of order $n \geq 3$ whose minimum degree satisfies $\delta(G) \leq \frac{\bar{d}(G)}{2}$, and suppose $\mathscr{LE}_G(w) < 0$ for some $w \in V(G)$. Then*

$$\max_{u,v \in V(G)} \left|\mathscr{LE}_G(u) - \mathscr{LE}_G(v)\right| \;\geq\; \left|\mathscr{LE}_G(w)\right| \;>\; 0,$$

and consequently $\mathrm{Var}(\mathcal{M}_L(G)) > 0$, so G is not Laplacian deck-homogeneous.

Proof. Let u be a vertex of minimum degree. Since $d_u = \delta(G) \leq \bar{d}(G)/2$, Theorem 3.4 gives $\mathscr{LE}_G(u) \geq 0$, whence

$$\begin{aligned}\max_{x,y} |\mathscr{LE}_G(x) - \mathscr{LE}_G(y)| &\geq \mathscr{LE}_G(u) - \mathscr{LE}_G(w) \\ &\geq -\mathscr{LE}_G(w) = |\mathscr{LE}_G(w)| > 0.\end{aligned}$$

A multiset with two distinct elements has strictly positive variance. □

The hypothesis of Corollary 5.6 is not vacuous, since the connected graphs on seven vertices with `graph6` strings `F~AGO` and `F?\vg` both have $\delta(G) = 1 \leq \bar{d}(G)/2$ and carry a vertex of negative local energy, hence a strictly positive profile variance.

5.2. **Separating Laplacian Cospectral Graphs, and a Census through Order Ten.** The global Laplacian energy $LE(G)$ is a function of $\mathrm{spec}(L(G))$ alone and is therefore constant on Laplacian cospectral classes. The profile, by contrast, is assembled from the vertex-deleted subgraphs $G^{(v)}$, which the spectrum does not determine. Since cospectral graphs have equal LE, the last assertion of Theorem 2.4 applies with $\tau = 0$: *two Laplacian cospectral graphs have the same profile if and only if they have the same vertex-deleted energy deck.* The profile therefore separates a cospectral pair exactly when some vertex-deleted energy occurs with different multiplicity in the two decks, a condition to which LE is blind. The three examples below show that both outcomes occur.

Example 5.7 (The smallest Laplacian cospectral pair)**.** The smallest pair of non-isomorphic connected Laplacian cospectral graphs has $n = 6$ and $m = 7$. The graph G_1 consists of a triangle and a 4-cycle sharing a vertex, and G_2 is $K_{2,3}$ with a pendant vertex attached to a vertex of degree 2. Both have $\varphi_L(x) = x(x-2)(x-3)^2(x^2-6x+4)$, hence $\mathrm{spec}(L) = \{0, 3-\sqrt{5}, 2, 3, 3, 3+\sqrt{5}\}$ and $LE(G_1) = LE(G_2) = 4 + 2\sqrt{5}$. Their decks differ, so by Theorem 2.4 so do their profiles

$$\begin{aligned}\mathcal{M}_L(G_1) &\approx \{0.4721,\ 1.5097^{(2)},\ 2.1320^{(2)},\ 3.2721\},\\ \mathcal{M}_L(G_2) &\approx \{1.9097^{(2)},\ 2.0721^{(2)},\ 2.1320^{(2)}\}.\end{aligned}$$

Here e_L already separates them (11.0277 against 12.2277), and the variances differ by almost two orders of magnitude (0.7184 against 0.0088), reflecting the hub of G_1 against the nearly uniform bipartite core of G_2.

The variance is a convenient summary, but it is strictly weaker than the profile, and it fails precisely on the most symmetric cospectral families.

Example 5.8 (Cospectral vertex-transitive graphs)**.** The 4×4 rook's graph R and the Shrikhande graph S are non-isomorphic vertex-transitive members of $\mathrm{srg}(16, 6, 2, 2)$, hence Laplacian cospectral with $\mathrm{spec}(L) = \{0, 4^{(6)}, 8^{(9)}\}$ and $LE = 36$. Their profiles are constant, so $\mathrm{Var}(\mathcal{M}_L(R)) = \mathrm{Var}(\mathcal{M}_L(S)) = 0$ by Remark 4.4, and the variance says nothing about the pair. The profiles themselves differ,

$$\mathcal{M}_L(R) = \{3.177795^{(16)}\}, \qquad \mathcal{M}_L(S) = \{3.223746^{(16)}\},$$

exactly as Remark 4.9 predicts from the first subconstituents $2K_3$ and C_6.

Example 5.9 (Cospectral graphs with identical profiles)**.** The graphs of order 9 and size 26 with `graph6` strings `HC~vfr~` and `H?~vvj~` are non-isomorphic, have the same degree sequence $(8, 6^{(4)}, 5^{(4)})$, and have the same integral Laplacian spectrum $\{0, 5^{(4)}, 7^{(2)}, 9^{(2)}\}$, so that $LE = \frac{160}{9}$ for both. Their vertex-deleted Laplacian spectra do not agree as multisets, yet their energy decks coincide, $\mathcal{D}_L = \{13, 14^{(4)}, 16^{(4)}\}$, and therefore so do their profiles,

$$\mathcal{M}_L = \left\{ \left(\tfrac{16}{9}\right)^{(4)}, \left(\tfrac{34}{9}\right)^{(4)}, \tfrac{43}{9} \right\}.$$

The census below shows that such pairs first occur at order nine, where there are six of them, and that there are seven at order ten.

Remark 5.10 (Scope of the profile). Examples 5.7 and 5.8 show that cospectrality does not force equal profiles, so $\mathcal{M}_L$ is strictly finer than LE. Example 5.9 shows that it is not a complete invariant, even among cospectral graphs, and that equal energy decks do not require equal spectral decks. The variance, finally, is no substitute for the profile, since it vanishes on every vertex-transitive graph.

An exhaustive census through order ten.

How rigid is simultaneous equienergeticity in practice, and how much of it survives the passage from e_L to the full profile? To answer this we classified all connected graphs of order $n \le 10$ under the equivalence relation

$$G \sim_L H \text{ if and only if } LE(G) = LE(H) \quad \text{and} \quad e_L(G) = e_L(H).$$

By Lemma 2.1(3), $G \sim_L H$ is equivalent to the simultaneous equality of the global energy and the mean deletion-deck energy ($LE(G) = LE(H)$ and $\overline{D}_L(G) = \overline{D}_L(H)$).

TABLE 2. Exhaustive classification of connected graphs under $\sim_L$. *Max* is the size of the largest class. The last four columns count, among all unordered $\sim_L$-pairs of the given order, those whose two graphs have equal size, equal degree sequence $\mathbf{d}$, equal Laplacian spectrum, and equal local profile.

n	Classes	Graphs	Pairs	Max	$= m$	$= \mathbf{d}$	Cosp.	$= \mathcal{M}_L$
6	3	7	5	3	0	0	0	1
7	20	41	22	3	15	5	2	1
8	135	305	258	10	85	47	37	6
9	1059	2296	1499	6	1080	642	596	53
10	17856	37296	24335	42	15852	12900	12335	19

Non-trivial $\sim_L$-classes emerge at order six and proliferate to 17856 classes covering 37296 graphs at order ten, the largest class $\mathcal{C}_{10,425}$ containing 42 mutually equivalent graphs. Table 2 also records, for each order, how many $\sim_L$-pairs agree in size, degree sequence, Laplacian spectrum and local profile. Three readings stand out.

(1) **Independence from the degree data.** Simultaneous energy equality forces neither equal size nor equal degree sequence. At order six all five $\sim_L$-pairs have distinct sizes and non-isomorphic degree sequences, and even at order ten some 35% of the pairs (8483 of them) have different sizes.

(2) **Partial overlap with cospectrality.** Laplacian cospectrality forces $LE(G) = LE(H)$, but it accounts for only about half of the pairs at order ten (12335 of 24335). The remaining 12000 pairs are Laplacian-noncospectral and yet share both first-order invariants.

(3) **Selectivity of the full profile.** Strong local equienergeticity is far more restrictive than equality of e_L. Only 19 of the 24335 pairs at order ten, that is 0.078%, preserve the whole multiset $\mathcal{M}_L$. As a proportion of $\sim_L$-pairs this last column is not monotone, falling from 3.54% at order nine to 0.078% at order ten. The entire order-ten row was recomputed independently [10] and all eight entries were reproduced, so the non-monotonicity is a property of the data and not of the enumeration. Part of it is explained by the constraint $\mathcal{M}_L(G) = \mathcal{M}_L(H)$ imposing n simultaneous real equalities against the two imposed by $G \sim_L H$; the size of the fall at order ten is not, and we record it as Problem 6.3.

The maximal class $\mathcal{C}_{10,425}$ illustrates all three readings at once. Its 42 graphs have ten distinct sizes ($22 \le m \le 34$), 42 distinct degree sequences and 42 distinct Laplacian characteristic polynomials, so no two of them are cospectral, and they realise 41 distinct profiles, a single pair sharing its profile.

Remark 5.11 (The coincidences are exact). A pair detected by comparing real numbers need not be a genuine coincidence, and equality of algebraic numbers cannot be decided by interval refinement, since equal numbers have overlapping enclosures at every precision. The 19 profile-preserving pairs at order ten were therefore certified by a symbolic evaluation of LE. Factor $\varphi_{L(F)} = \prod_i f_i^{e_i}$ over $\mathbb{Q}$ and let t_i be the number of roots of f_i above $\overline{d}(F)$, obtained exactly by a Sturm sequence. As in Lemma 3.2,

$$LE(F) = 2\Big(\sum_i e_i \Sigma_{f_i,t_i} - k^* \overline{d}(F)\Big), \qquad k^* = \sum_i e_i t_i, \tag{30}$$

where $\Sigma_{f,t}$ is the sum of the t largest roots of f. Factors with $t_i \in \{0, \deg f_i\}$ contribute rational amounts. A factor straddling the baseline contributes an irrational amount that depends only on (f_i, t_i), and even then only up to a rational shift and a reflection. With μ_f the mean root of f and $f_0(x) = f(x + \mu_f)$ one has $\Sigma_{f,t} = t\mu_f + \Sigma_{f_0,t}$ and $\Sigma_{f_0,t} = \Sigma_{\widehat{f}_0,\deg f - t}$ for $\widehat{f}_0(x) = (-1)^{\deg f} f_0(-x)$. Two graphs whose rational parts agree and whose normalised straddling pairs coincide as multisets therefore have equal LE; the test needs only integer factorisation, Sturm sequences and rational arithmetic. All 19 pairs pass it, as do their deletion decks, and so do all 80 profile-preserving pairs of orders six to ten. Five have rational profiles, for instance $\{(\frac{34}{15})^{(4)}, (\frac{18}{5})^{(2)}, (\frac{64}{15})^{(4)}\}$; in three of the remaining fourteen the two straddling cubics are reflections of one another, as with $f_1(x) = x^3 - 20x^2 + 122x - 212$ and $f_2(x) = -f_1(12 - x)$, whose partial root sums differ by exactly 8. The profile-preserving pairs at orders six and seven have the rational profiles $\{(\frac{16}{5})^{(3)}, (\frac{18}{5})^{(3)}\}$ and $\{(\frac{8}{7})^{(2)}, (\frac{22}{7})^{(4)}, \frac{36}{7}\}$, the latter from Example 2.8(b). The coincidences are exact algebraic identities, not numerical near-misses.

Computational protocol. All connected graphs were generated with `geng` [21], energies were computed in double precision, and graphs were grouped with tolerance 10^{-9}. Every statement drawn from these data was then confirmed exactly: Laplacian cospectrality through the integer power sums $\operatorname{tr} L^k$, and every profile equality through the symbolic evaluation of Remark 5.11. The grouping does not depend on the tolerance: at order ten the classes are the same for every tolerance between 7×10^{-13} and 2×10^{-9}, and the window is wider at lower orders. Code, data and logs are available in the accompanying repository [10].

6. Conclusion

In this paper we developed the local Laplacian energy $\mathscr{LE}_G(v) = LE(G) - LE(G^{(v)})$ and its multiset profile $\mathcal{M}_L(G)$ as spectral invariants of vertex deletion.

The first conclusion is structural. Lemma 2.1 shows that $LE(G)$, $e_L(G)$ and the mean deck energy $\overline{D}_L(G)$ form a rank-two system; Theorem 2.4 identifies equality of profiles with a rigid translation of the deletion deck; and Theorem 2.6 shows that this identification persists through

every central moment. Up to a single additive constant, therefore, the profile $\mathcal{M}_L(G)$ carries exactly the information of the vertex-deleted energy deck $\mathcal{D}_L(G)$, and every subsequent result of the paper is a statement about that deck.

We then settled when the deletion of a vertex can increase the Laplacian energy, a question with no counterpart for the adjacency energy, which deletion never increases. Lemma 3.1 isolates the baseline threshold $d_v = \overline{d}(G)/2$, and Theorem 3.4 shows that it is genuinely necessary: deleting a vertex with $d_v \le \overline{d}(G)/2$ never increases the Laplacian energy, so $\mathscr{LE}_G(v) < 0$ can occur only at vertices heavier than half the average degree. The complete graphs, the cycles of Corollary 4.11, and the conference graphs of Corollary 4.8 show that the condition is not sufficient.

The family evaluations of Section 4 each carry a structural reading. The local energy never exceeds $LE(G)$, with equality only at star centres, and reaches $-\frac{n}{2}+O(1)$ on $K_{r,r}$. For a strongly regular graph it is a function of the spectrum of the first subconstituent alone; for a cycle it is positive and tends to $4/\pi$; for $K_n \circ K_1$ the profile splits into two atoms whose separation grows linearly in the order, so that together with the vertex-transitive graphs the profile spread ranges from 0 to order N; and in a generalized windmill the hub and the blade vertices carry opposite signs, the total e_L recording which side prevails.

The profile differs from the uncentered deletion profile vertex by vertex by at most the baseline shift (Proposition 5.1), and at the second-moment level we bounded the discrepancy between $\mathrm{Var}(\mathcal{M}_L(G))$ and the variance of the uncentered deletion profile by the degree modulus $\kappa(G) = \sqrt{4\sigma_d^2 + \overline{d}(G)^2}$. Example 5.5 shows that the term $\overline{d}(G)^2$ cannot be removed, and the stars show that the modulus itself is asymptotically sharp. Theorem 2.4 then identifies exactly when the profile separates a Laplacian cospectral pair, namely when the two deletion decks differ, a condition to which LE is blind, and Example 5.9 shows that the separation can fail: the smallest cospectral pairs with identical profiles have nine vertices. Example 5.8 marks the limits of the variance as a summary, since it vanishes on every vertex-transitive graph and so cannot separate cospectral pairs drawn from that class, even when the full profile does. Finally, the census through order ten confirms the selectivity of the profile in practice, and the symbolic evaluation of Remark 5.11 turns each of the nineteen order-ten coincidences into a verifiable algebraic identity.

We close with the questions the present work leaves open.

Problem 6.1. Theorem 3.4 shows that $\mathscr{LE}_G(v) < 0$ forces $d_v > \overline{d}(G)/2$, and the complete graphs show that the converse fails. Characterize the pairs (G, v) with $\mathscr{LE}_G(v) < 0$.

Problem 6.2. For even $n \ge 4$, is the minimum of $\mathscr{LE}_G(v)$ over connected graphs G of order n and vertices $v \in V(G)$ attained by $K_{n/2,n/2}$, with the value $-\frac{n}{2} + \frac{5}{2} + \frac{1}{2(n-1)}$ of Eq. (15)? This holds for $n = 4, 6, 8, 10$, where $K_{n/2,n/2}$ is in each case the unique minimiser. For $n = 5, 7, 9$ the minima are approximately 0.8721 and -0.3110, and exactly $-\frac{5}{3}$, none of them attained by a complete bipartite graph. What is the minimum for odd n?

Problem 6.3. Explain the behaviour of the last column of Table 2. The proportion of $\sim_L$-pairs preserving the full profile falls from 3.54% at order nine to 0.078% at order ten, a drop that the elementary counting heuristic does not account for.

Beyond these, the profile invites study in graph reconstruction, network robustness and structural centrality, and the symbolic evaluation of Remark 5.11 may be of independent interest wherever energies of integer matrices have to be compared exactly.

Declaration of competing interest

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Data availability

The code and data supporting this study are openly available in the repository https://github.com/cahitdede/local-laplacian-energy [10], archived on Zenodo under doi:10.5281/zenodo.23298644. The repository reproduces every computational statement of the paper.

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